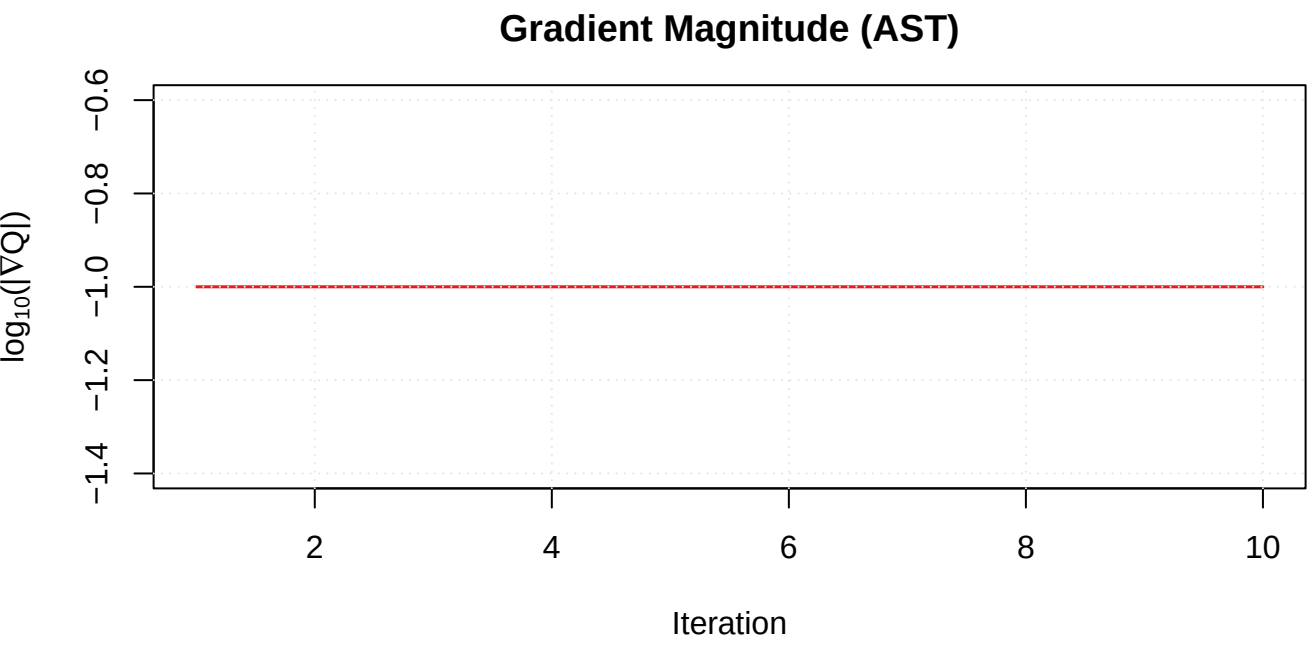
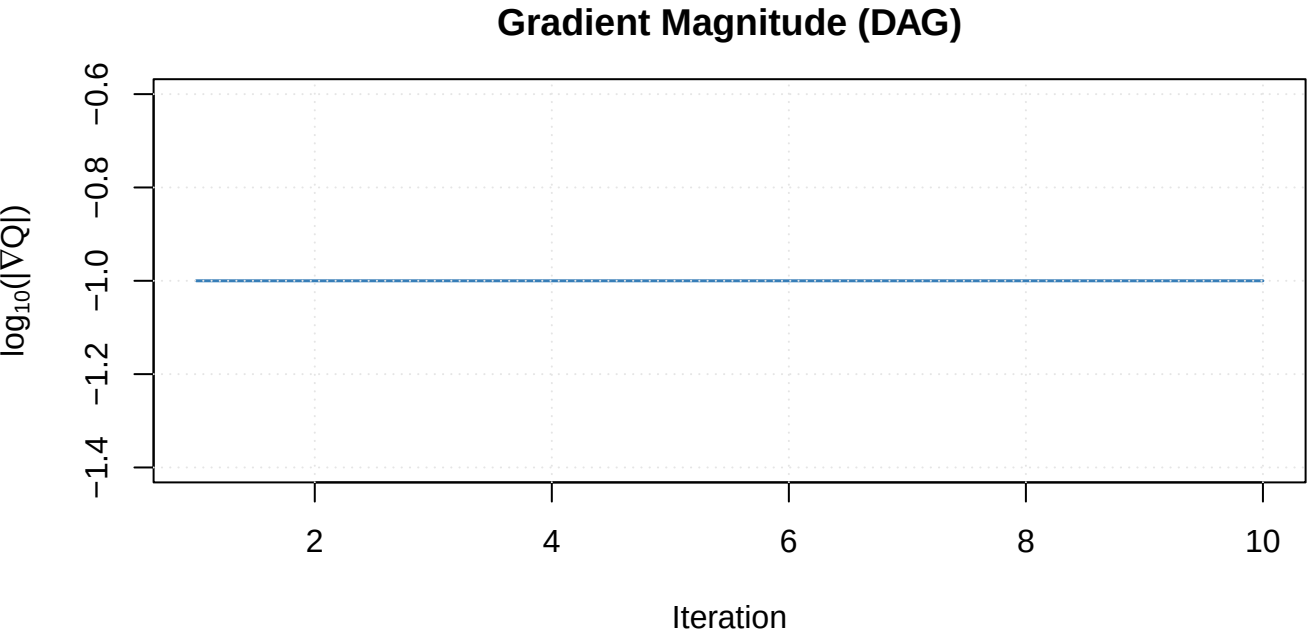

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration


Gradient Magnitude (DAG)
log10(|∇Q|)
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−0.8
−1.0
−1.2
−1.4
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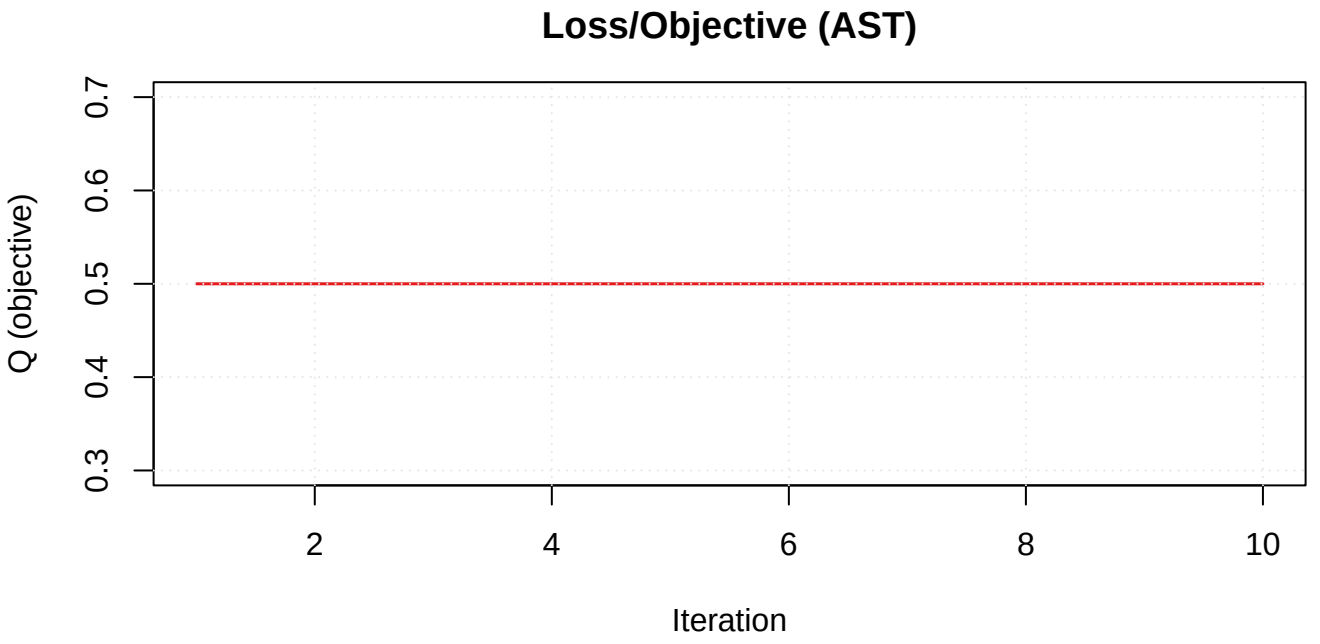
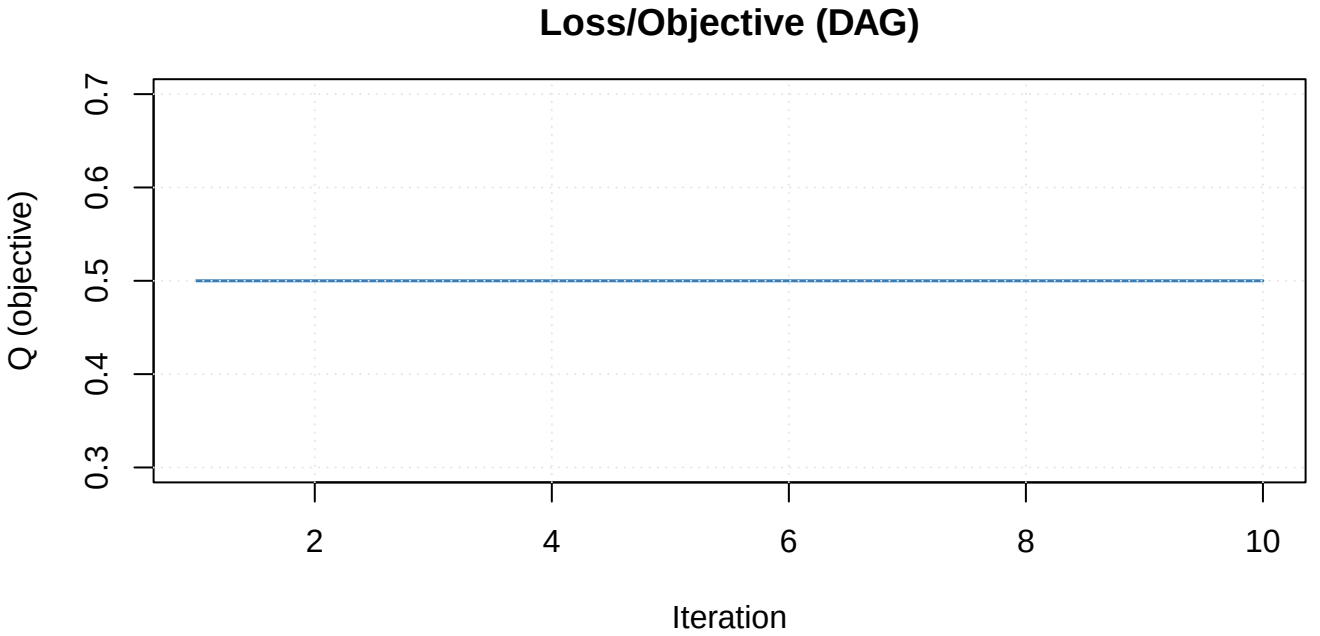
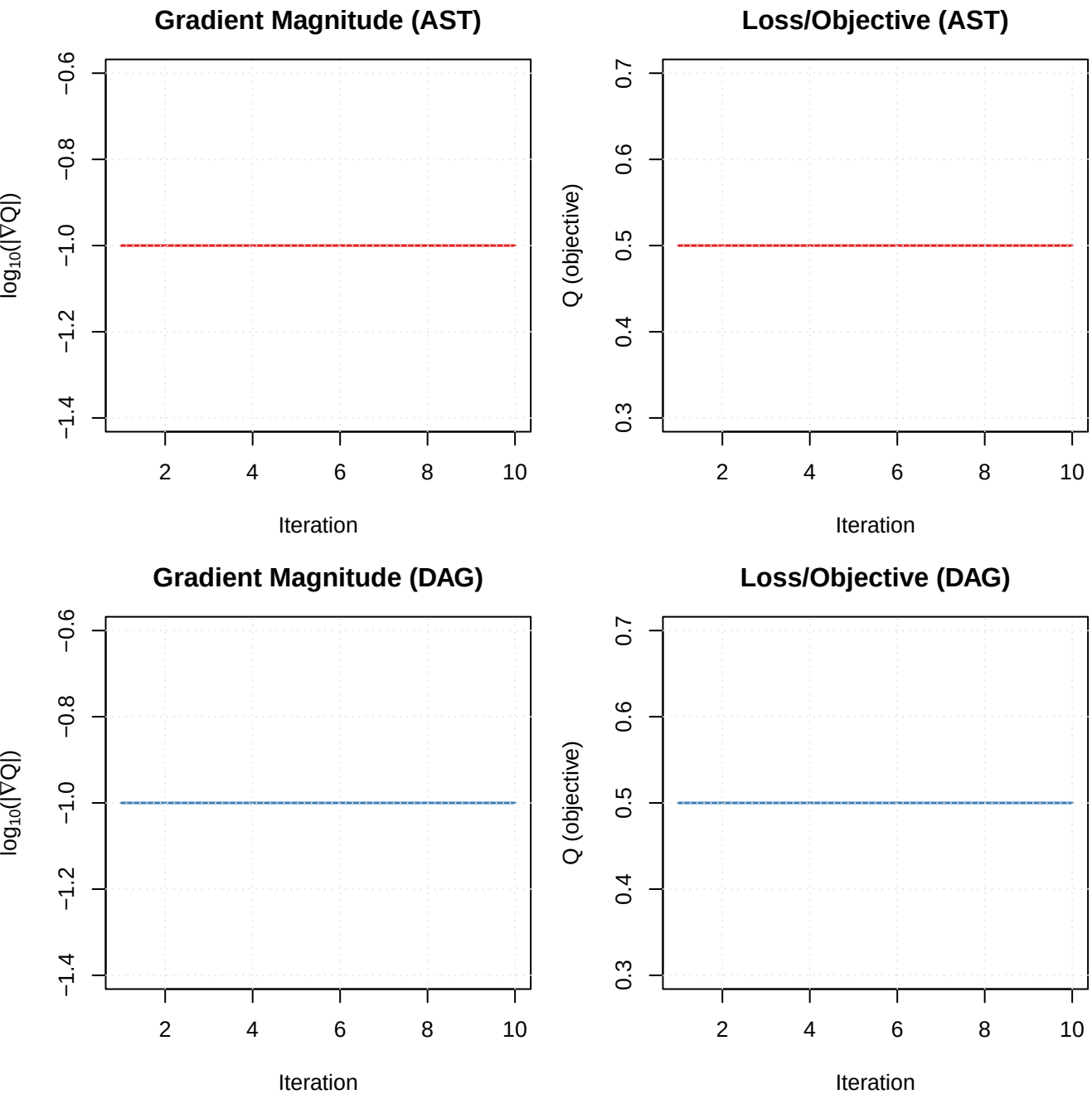
4
6
8
10
Iteration


Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration


Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration


Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
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0.4
0.3
2
4
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8
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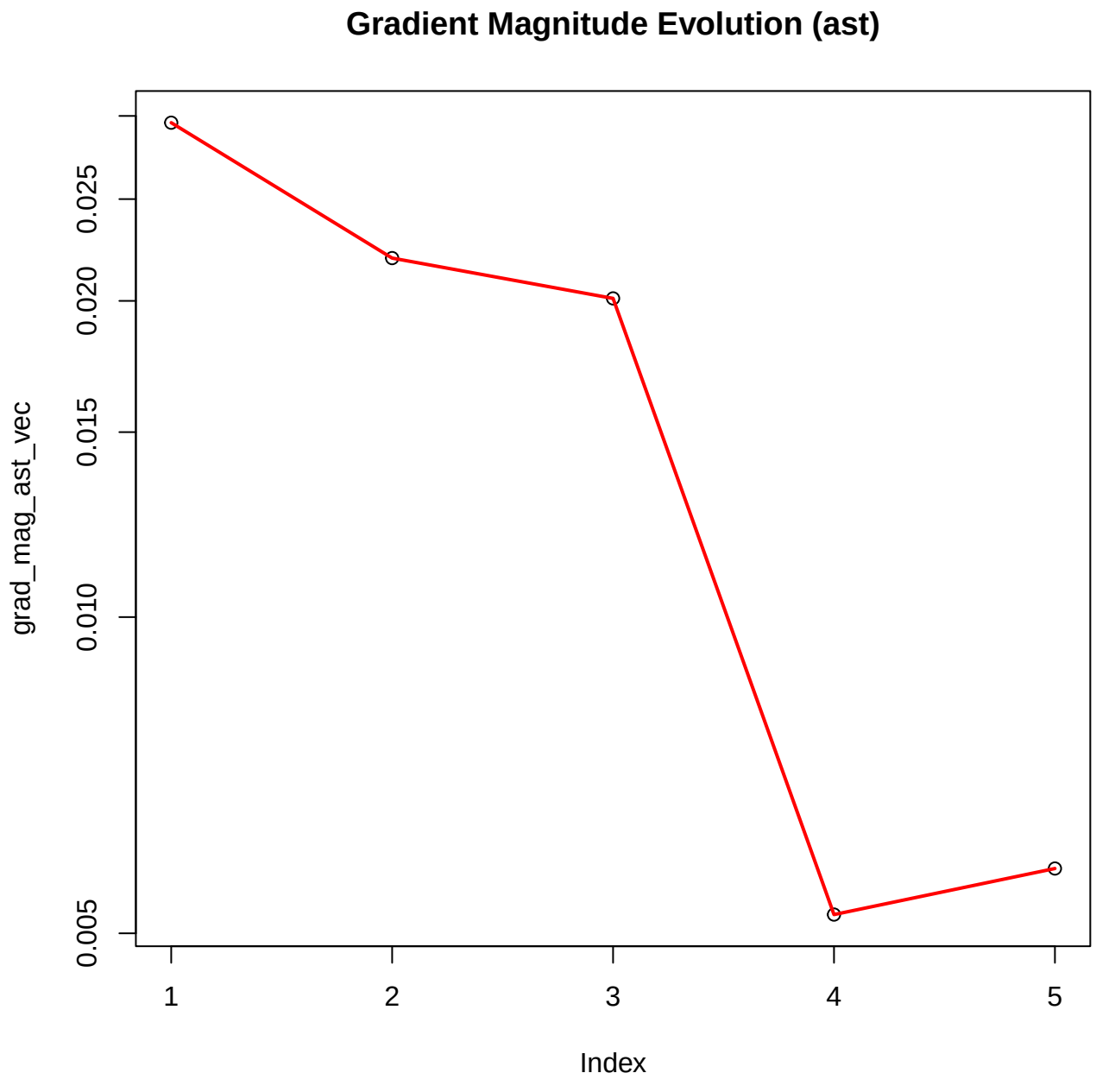
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
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3
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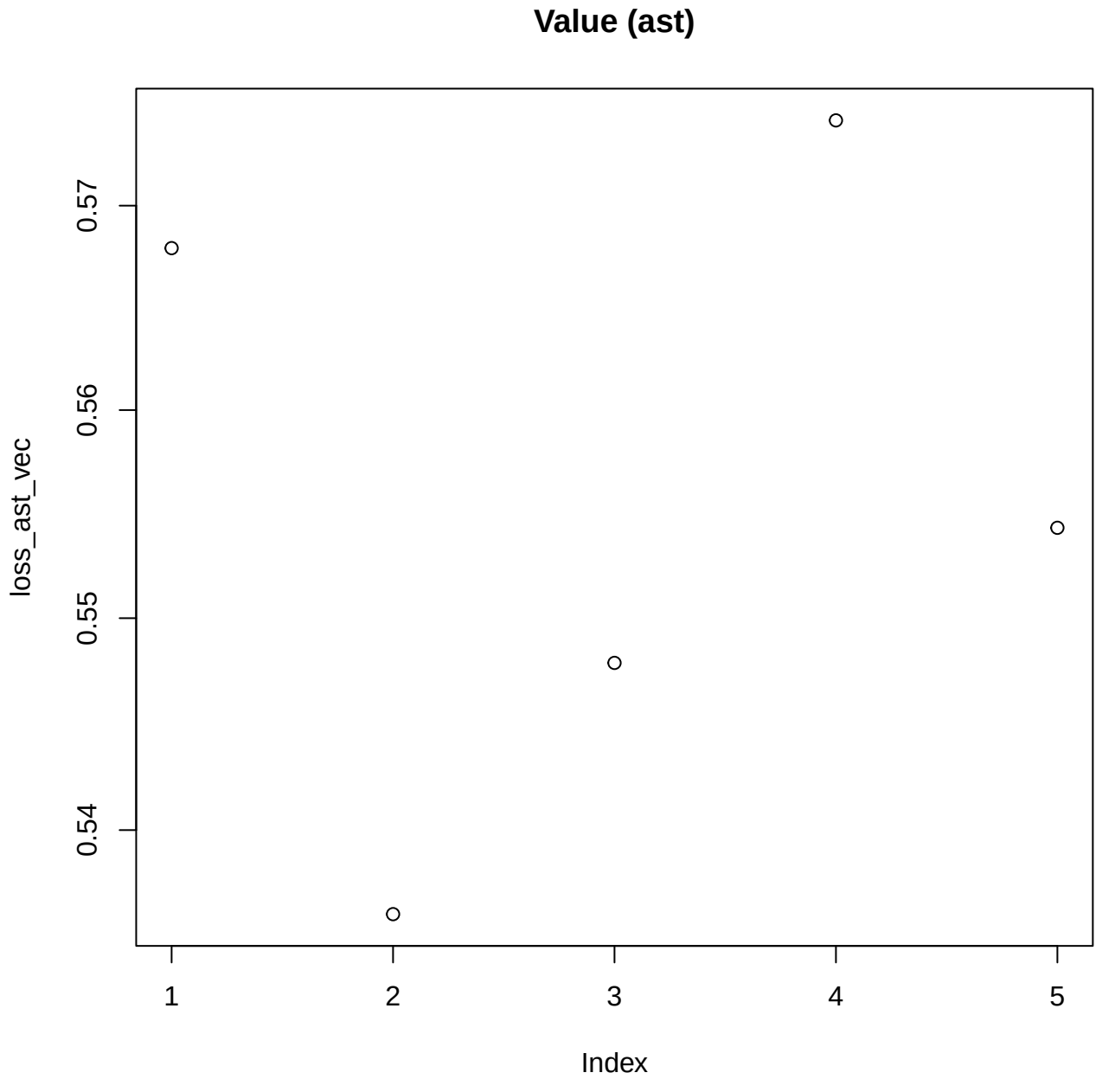
5
Index


Value (ast)
loss_ast_vec
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0.55
0.56
0.57
1
2
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5
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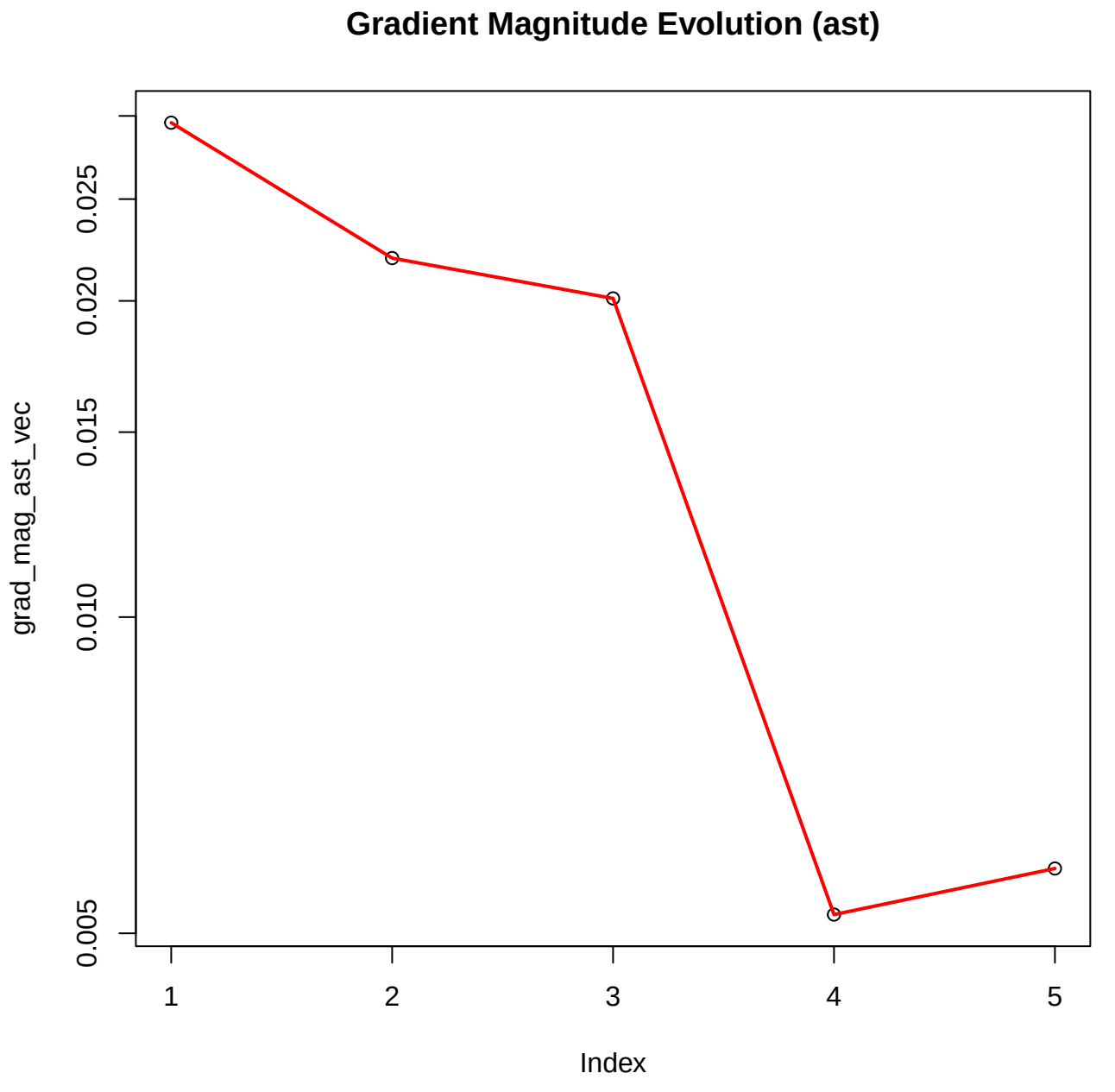
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.025
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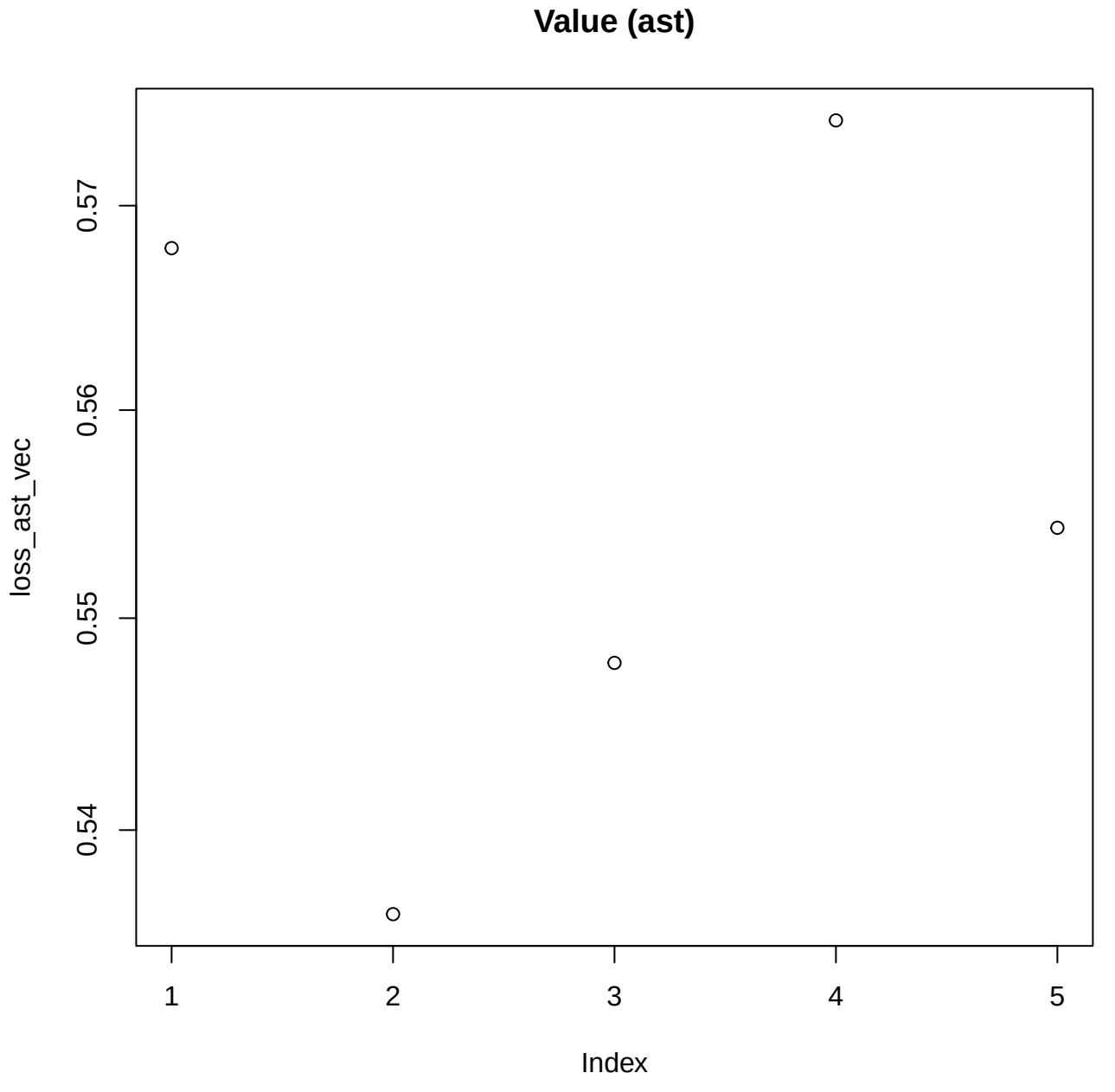
Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
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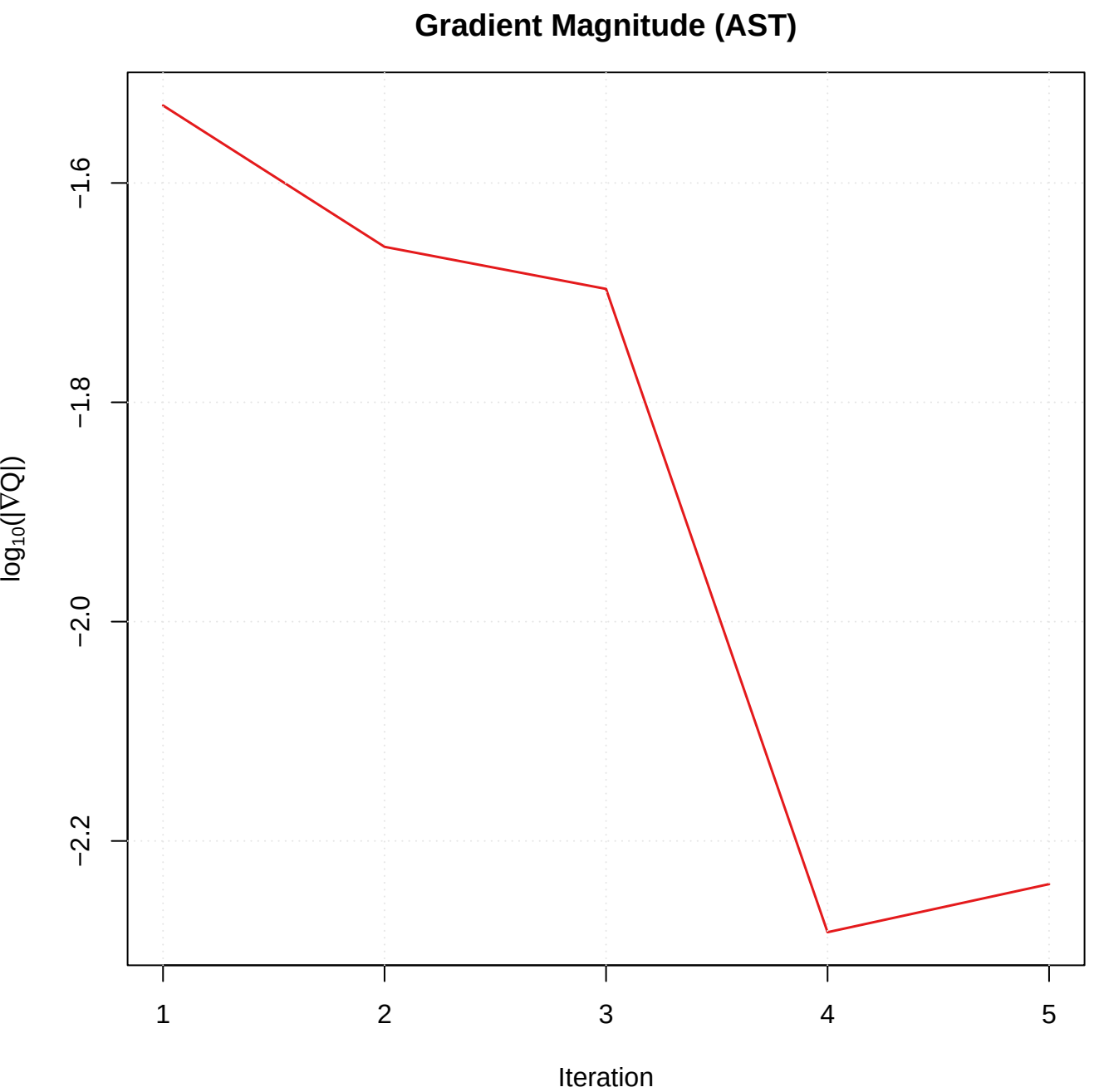
5
Index


Gradient Magnitude (AST)
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−1.8
−2.0
−2.2
log10(|∇Q|)
1
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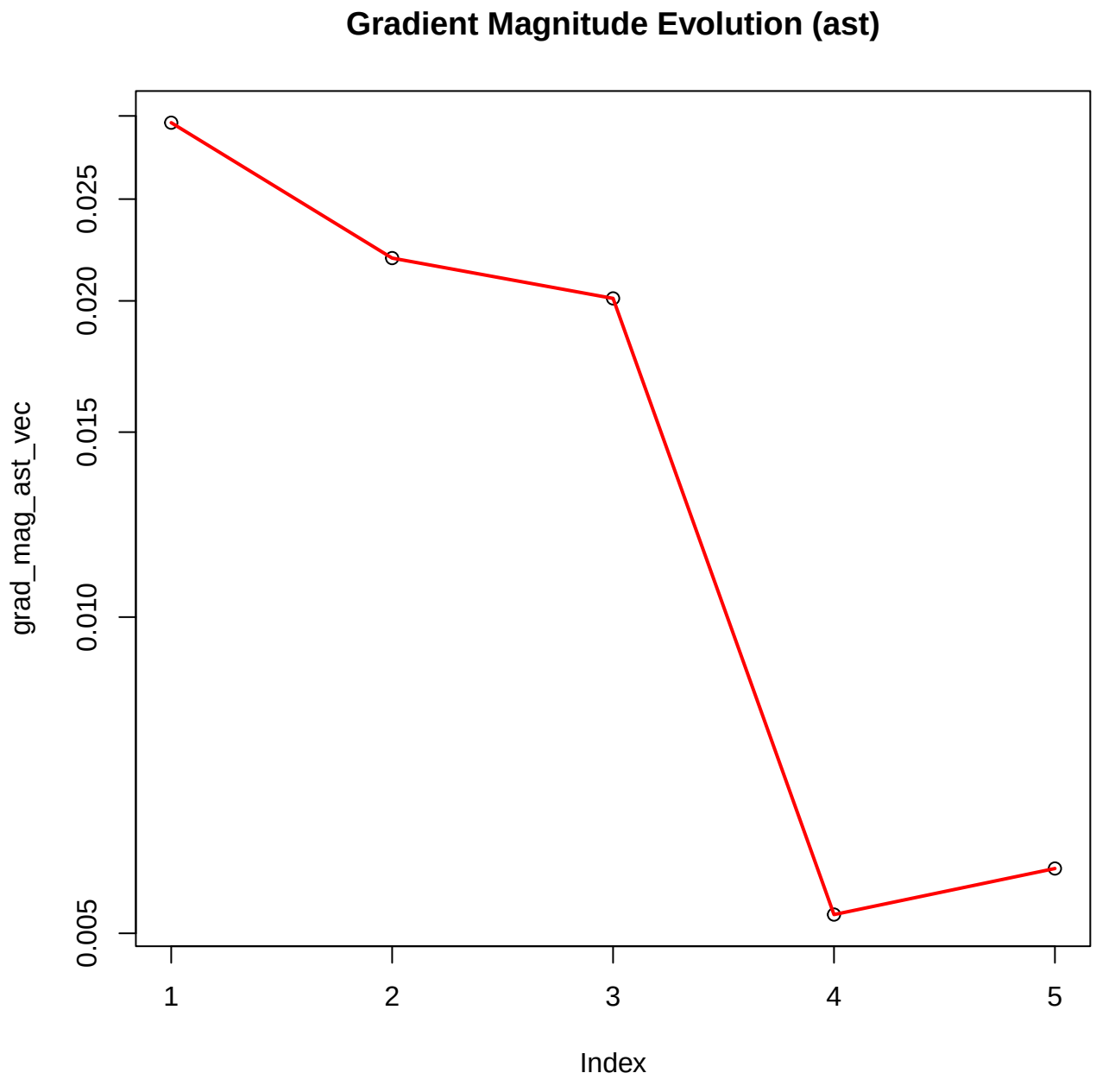
3
4
5
Iteration


Gradient Magnitude Evolution (ast)
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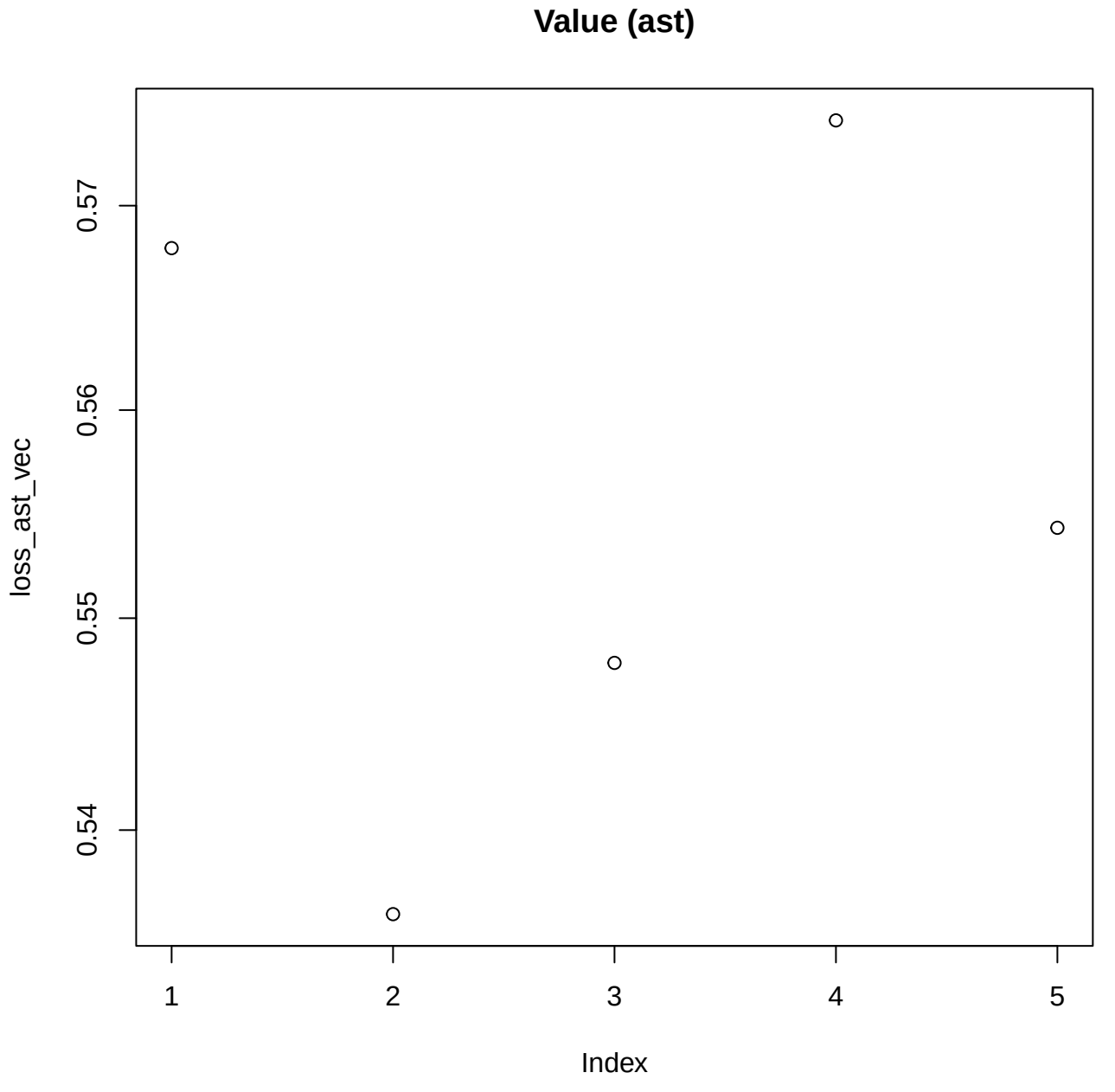
Value (ast)
loss_ast_vec
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0.55
0.56
0.57
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5
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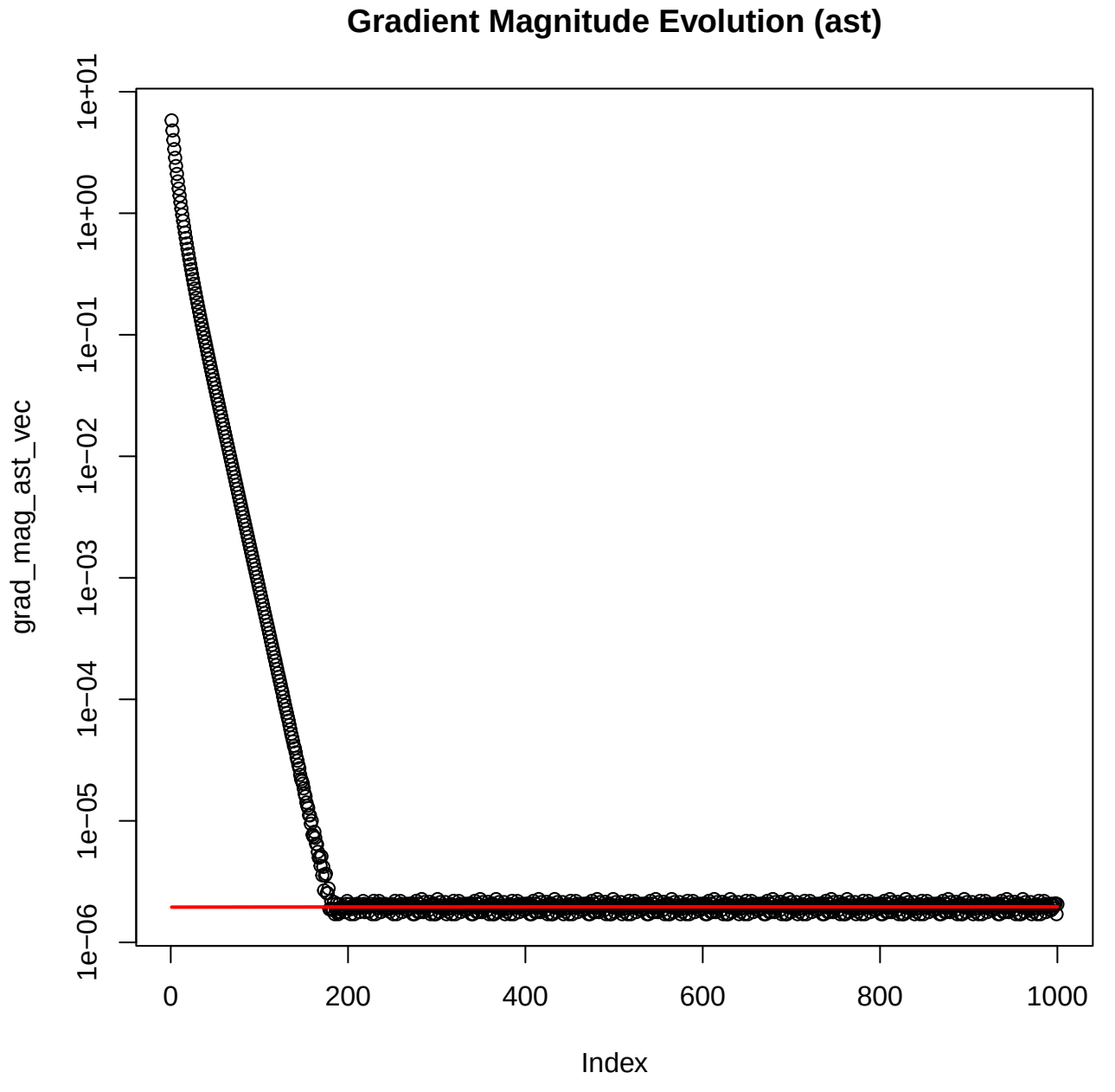
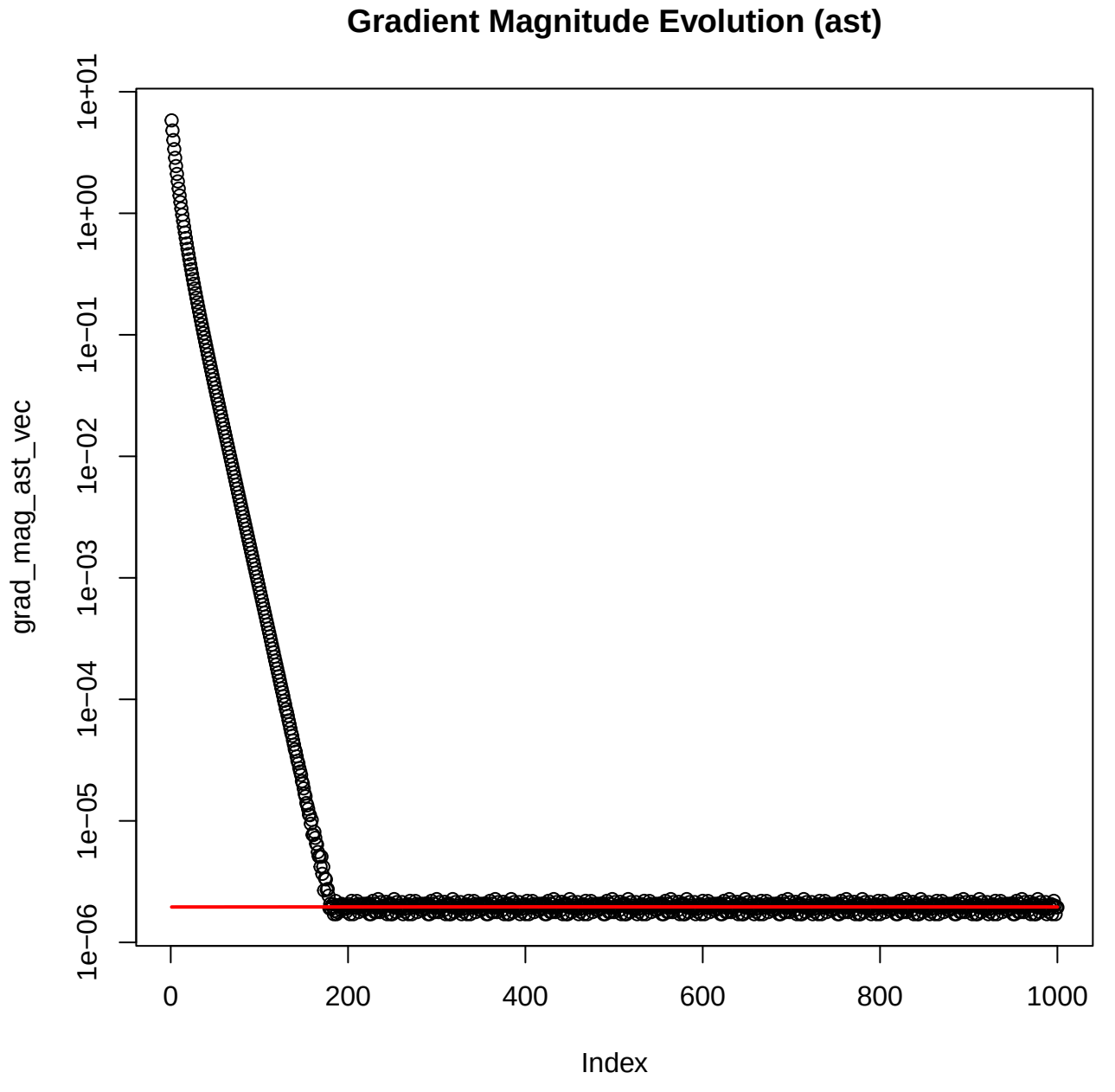

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
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600
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1000
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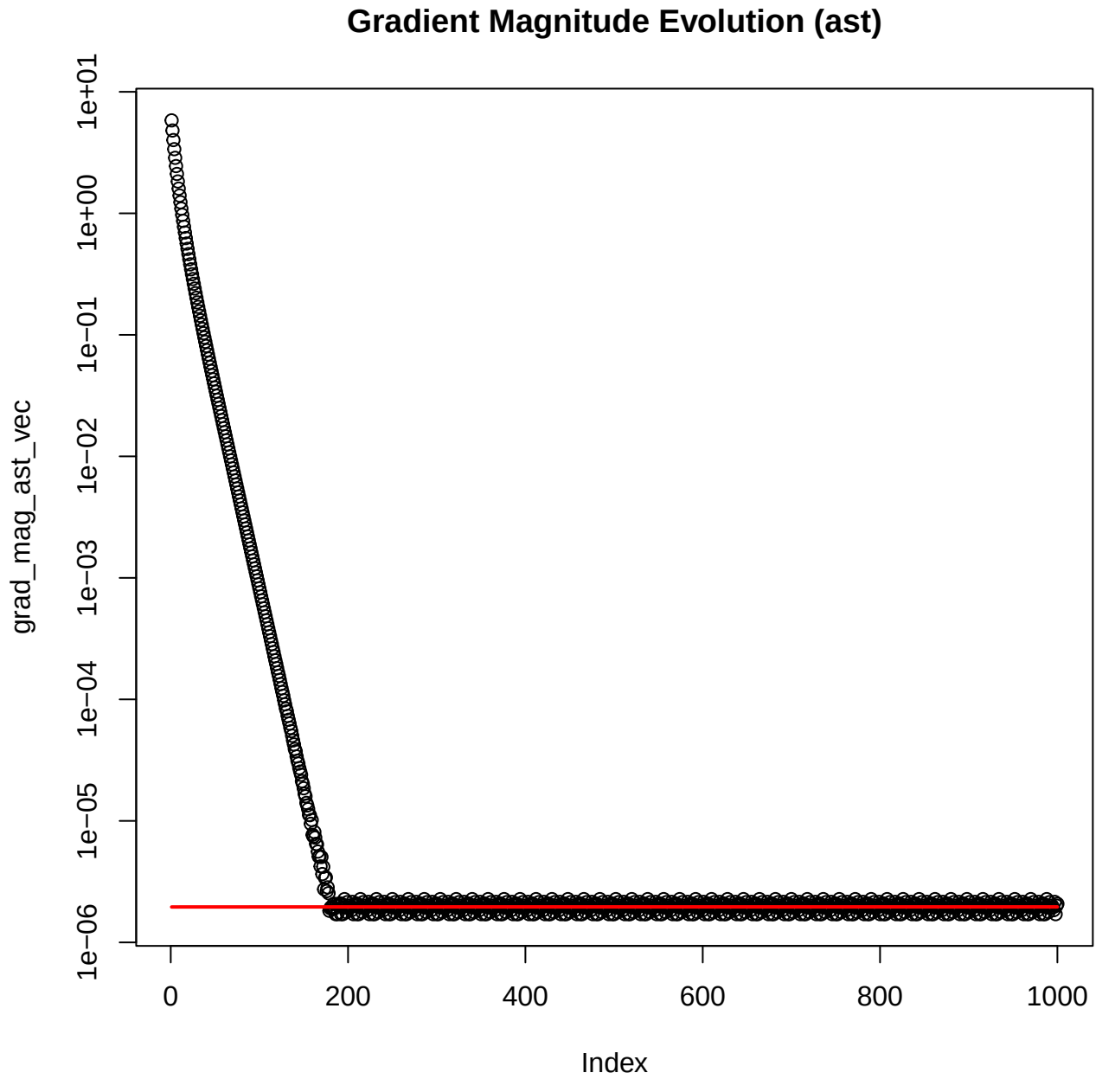
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
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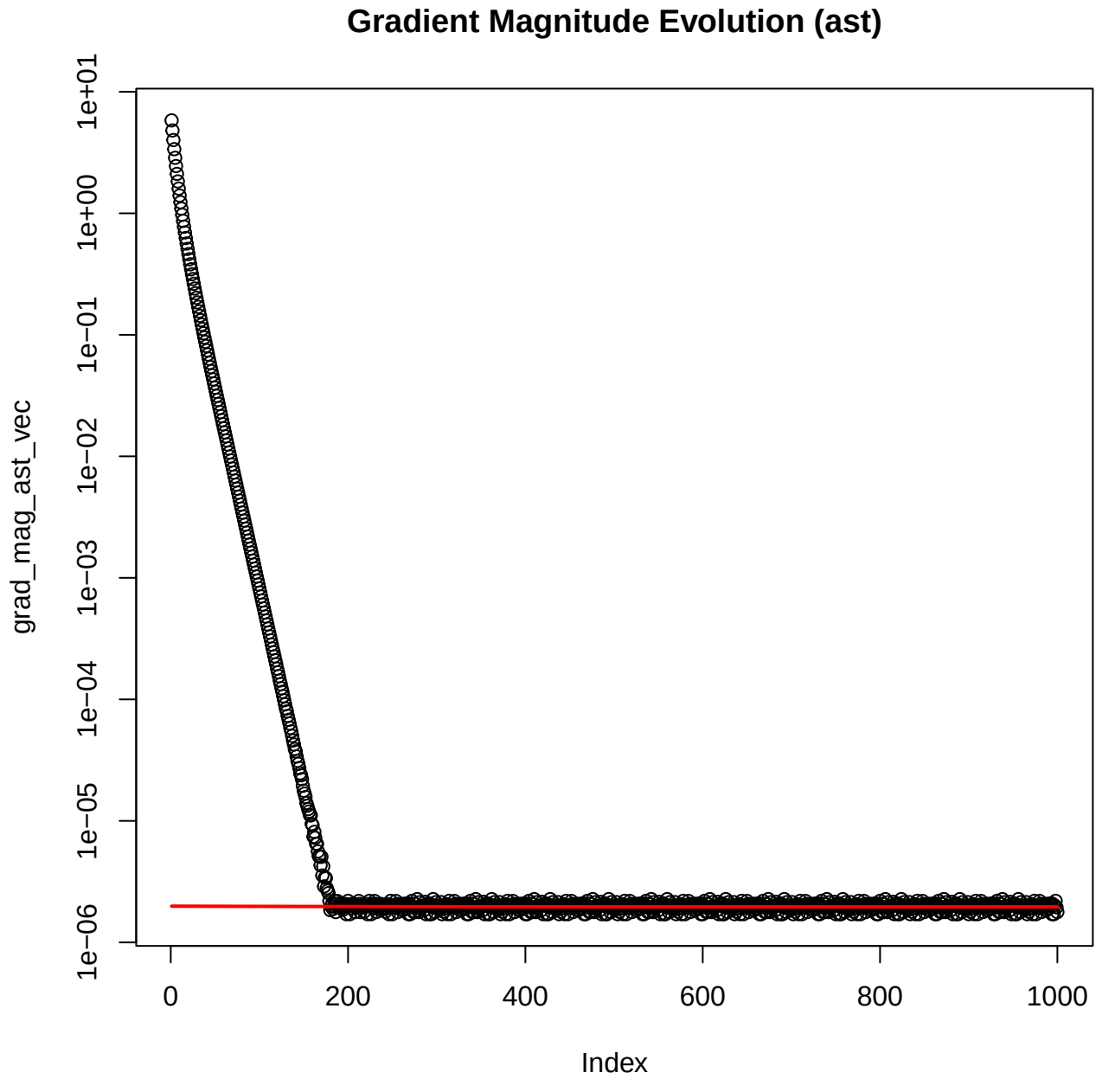
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
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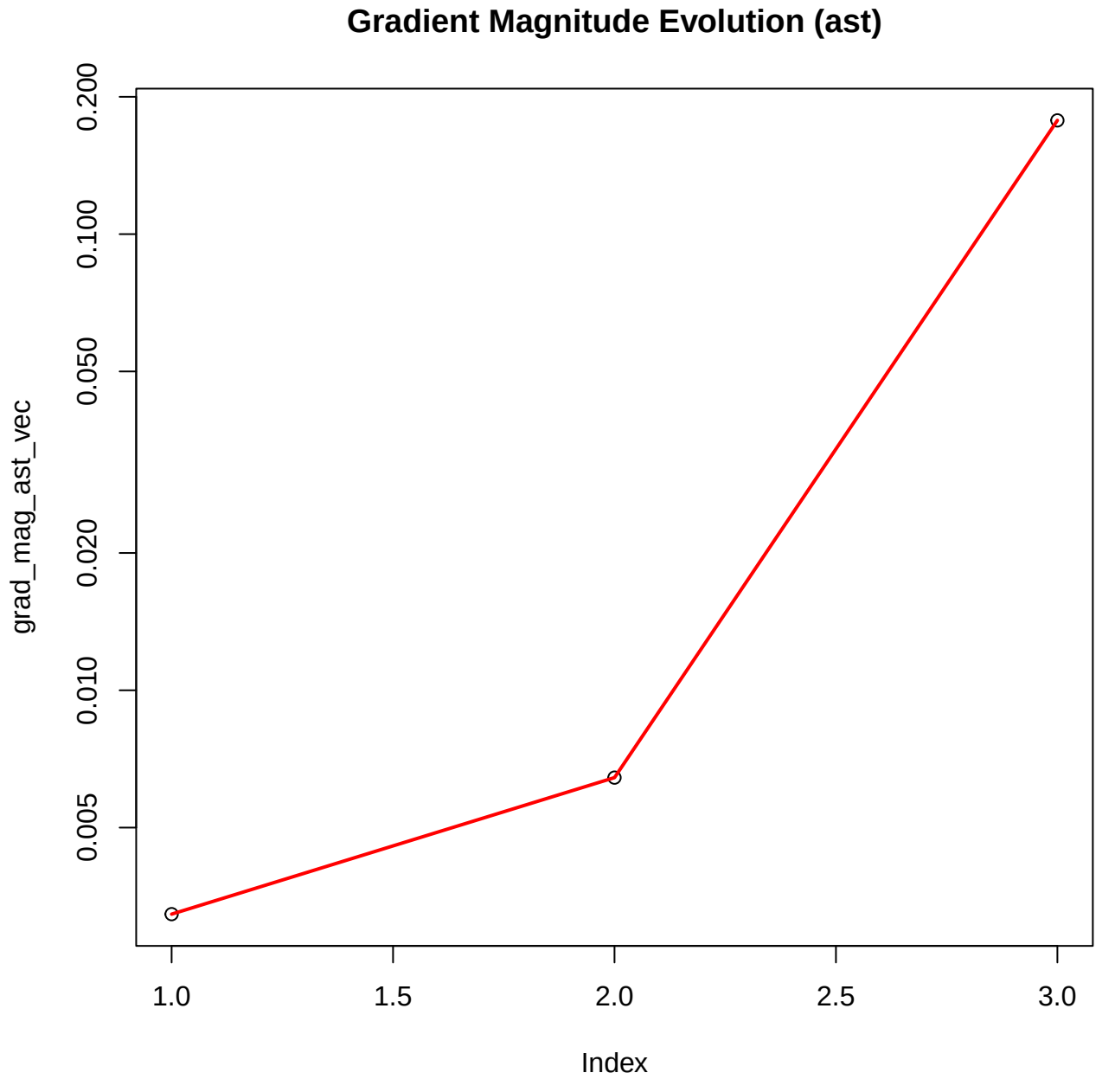
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.020
0.050
0.100
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2.0
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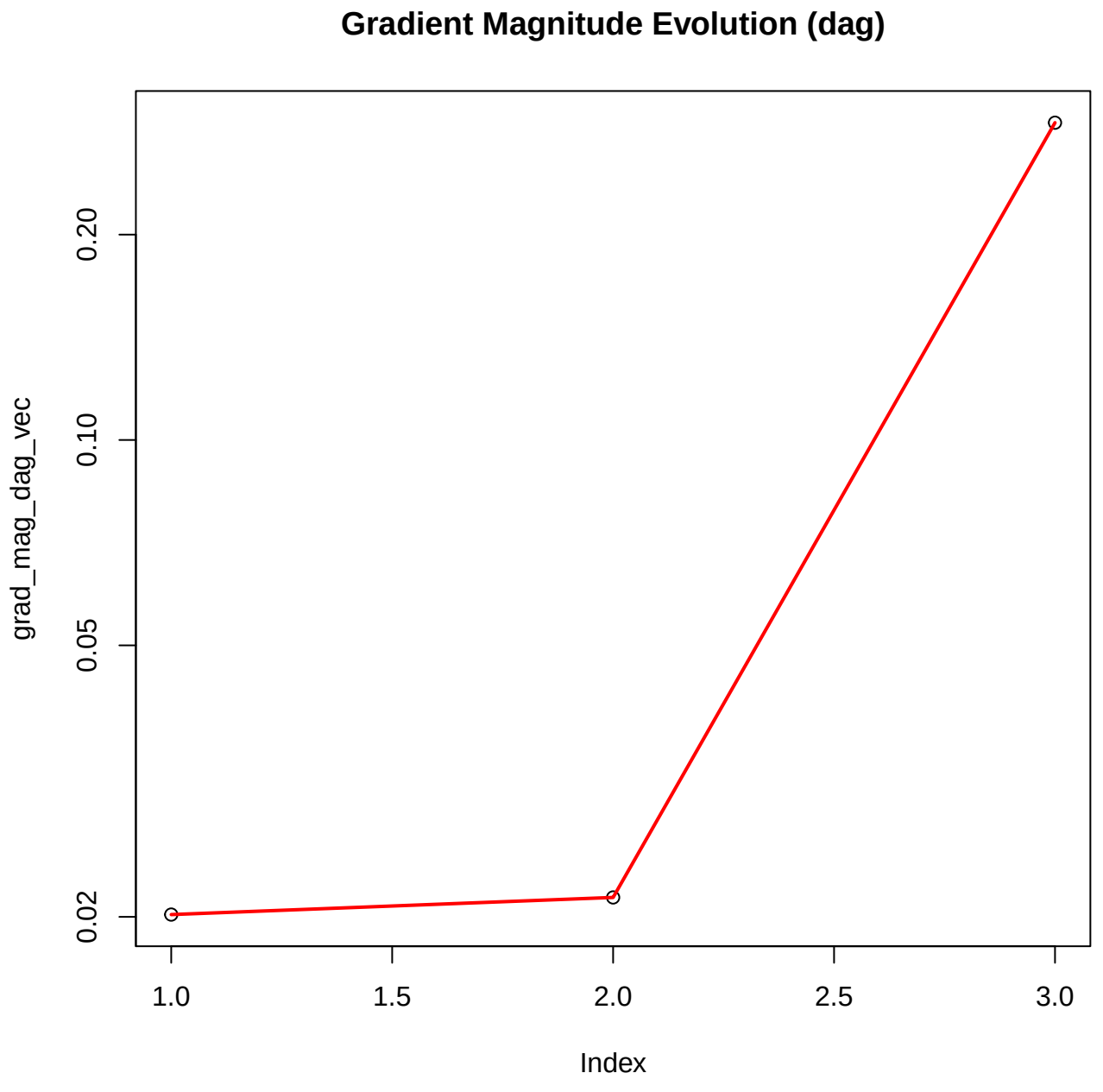
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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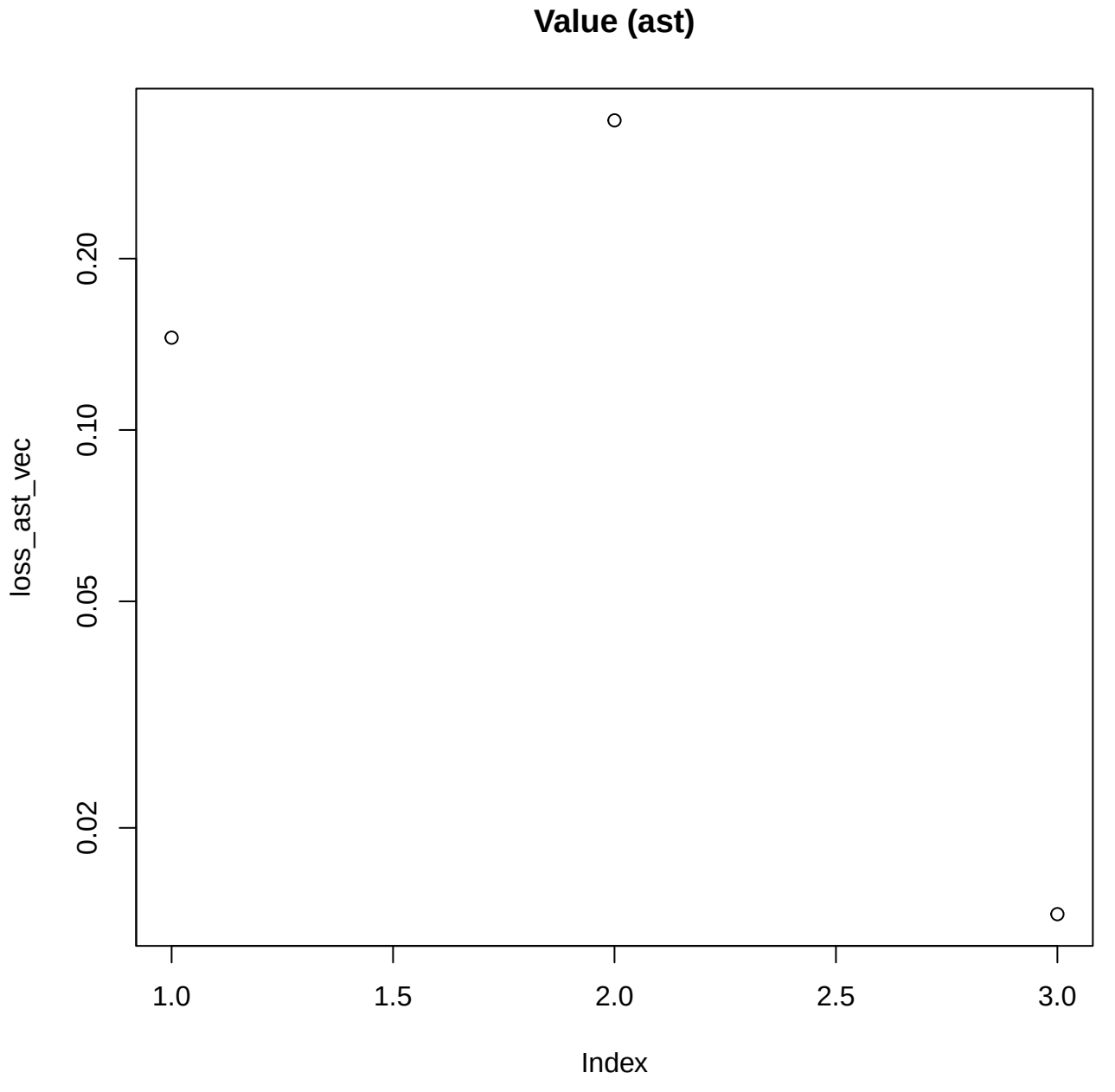
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index


Value (ast)
loss_ast_vec
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0.05
0.10
0.20
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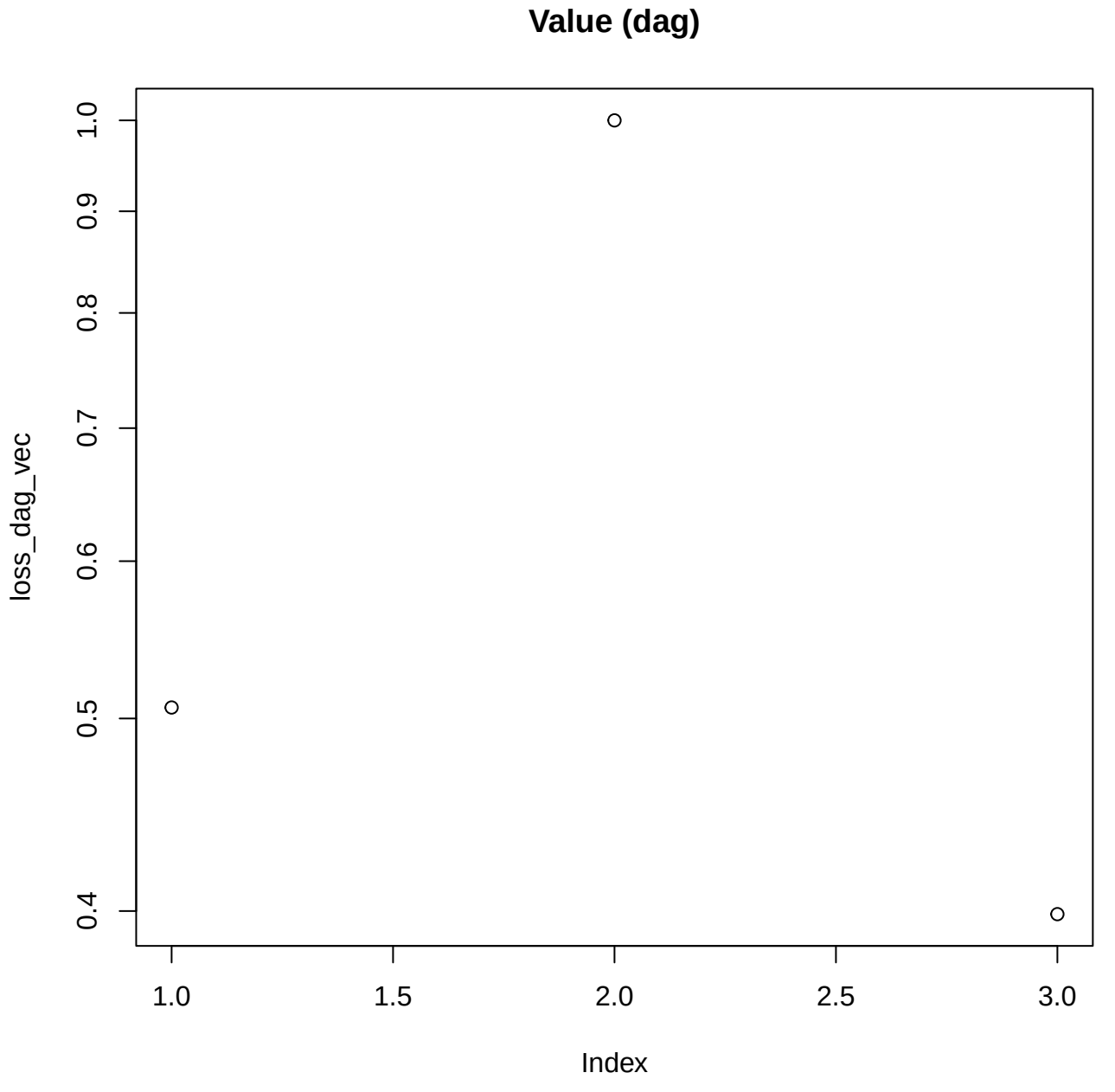
1.5
2.0
2.5
3.0
Index


Value (dag)
loss_dag_vec
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0.7
0.6
0.5
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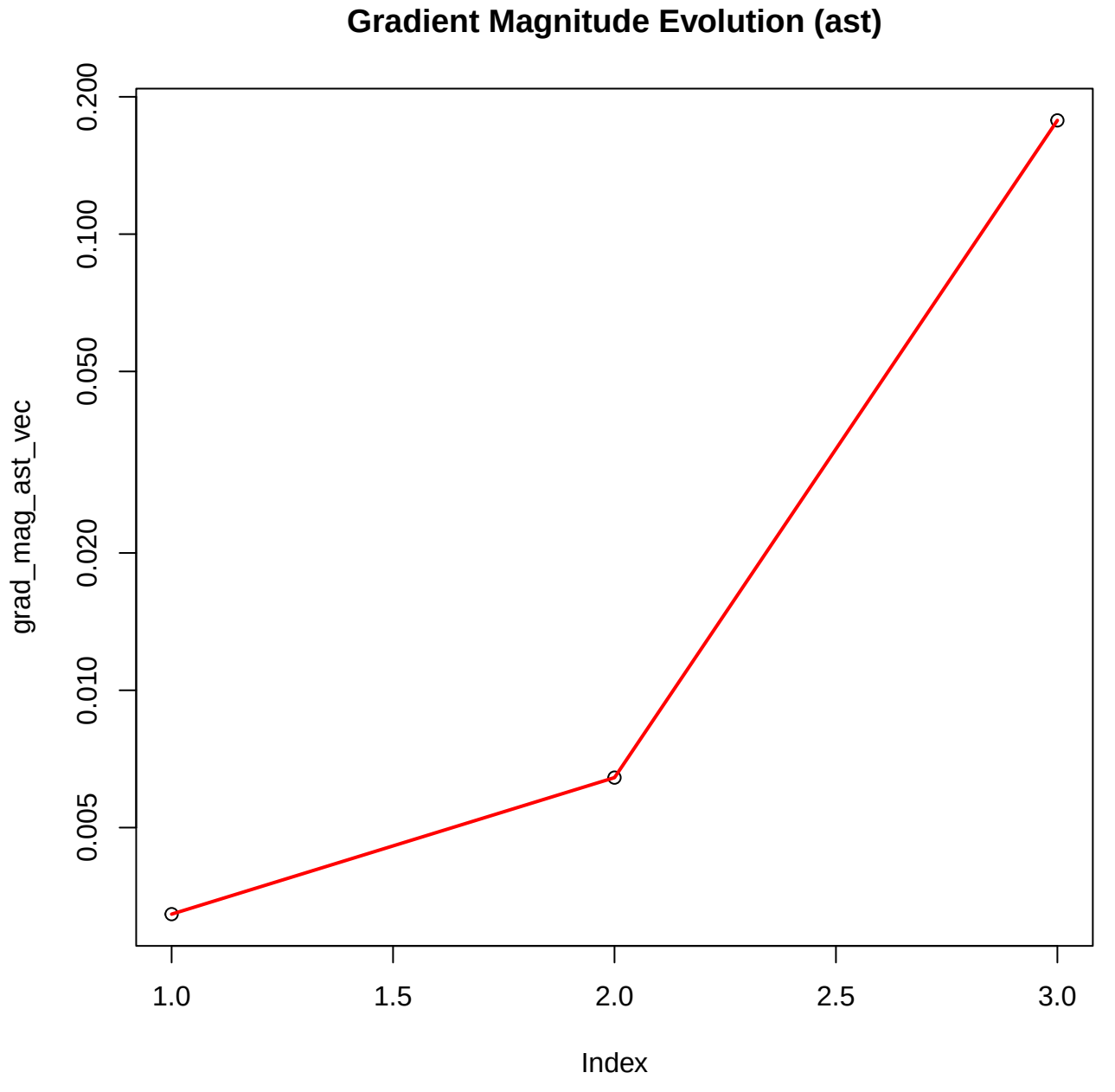
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
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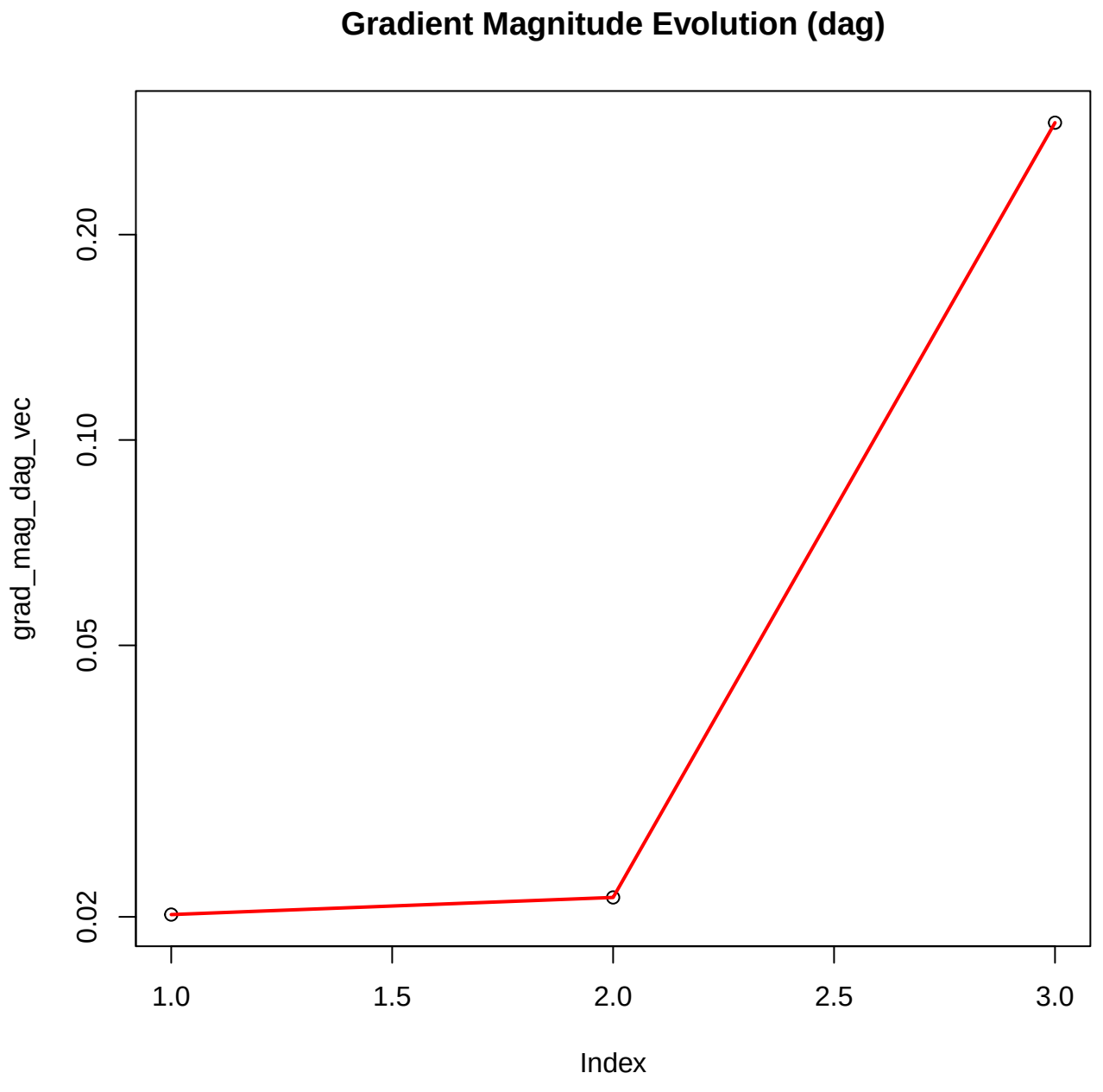
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.10
0.20
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1.5
2.0
2.5
3.0
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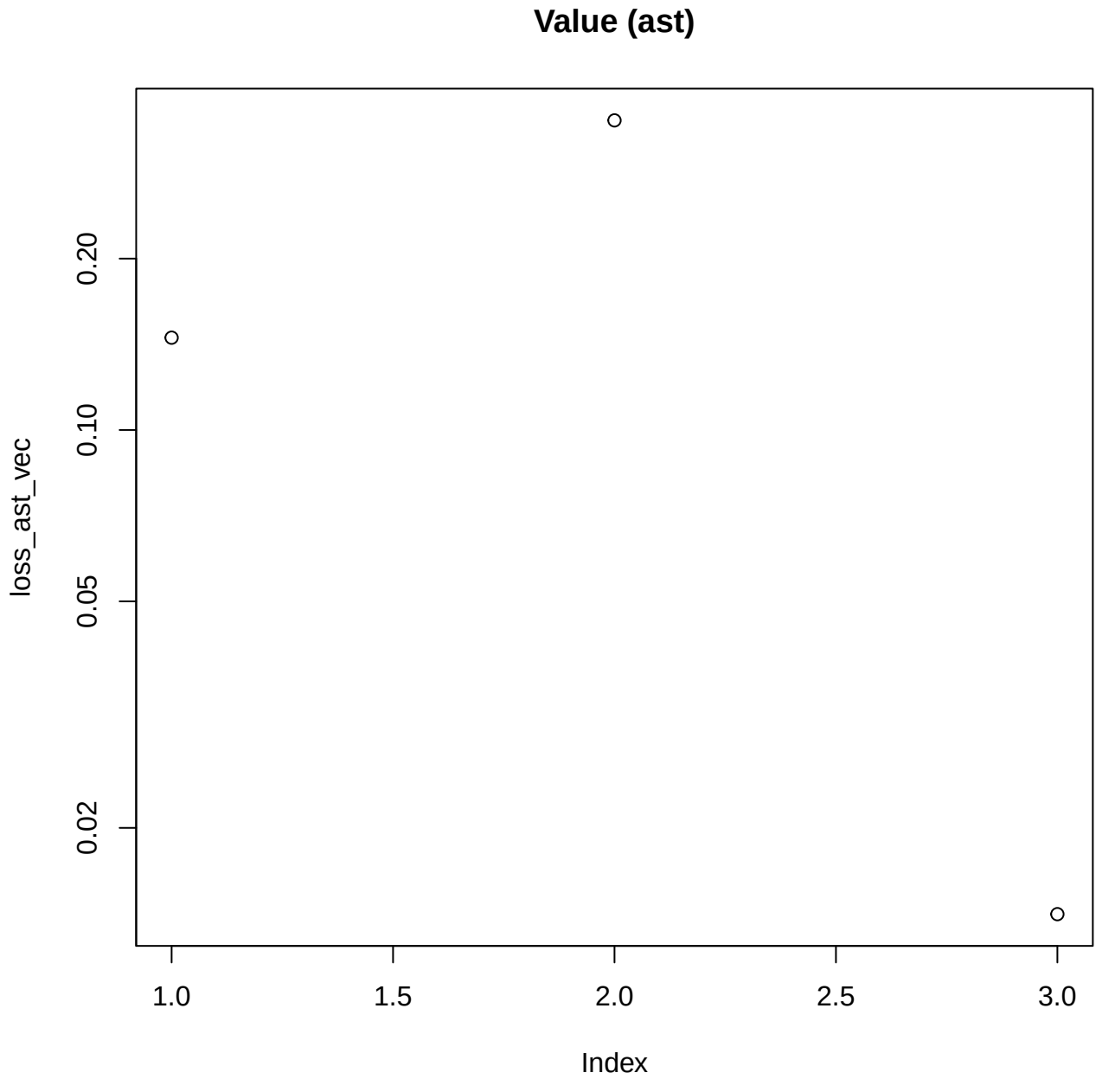
Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
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1.5
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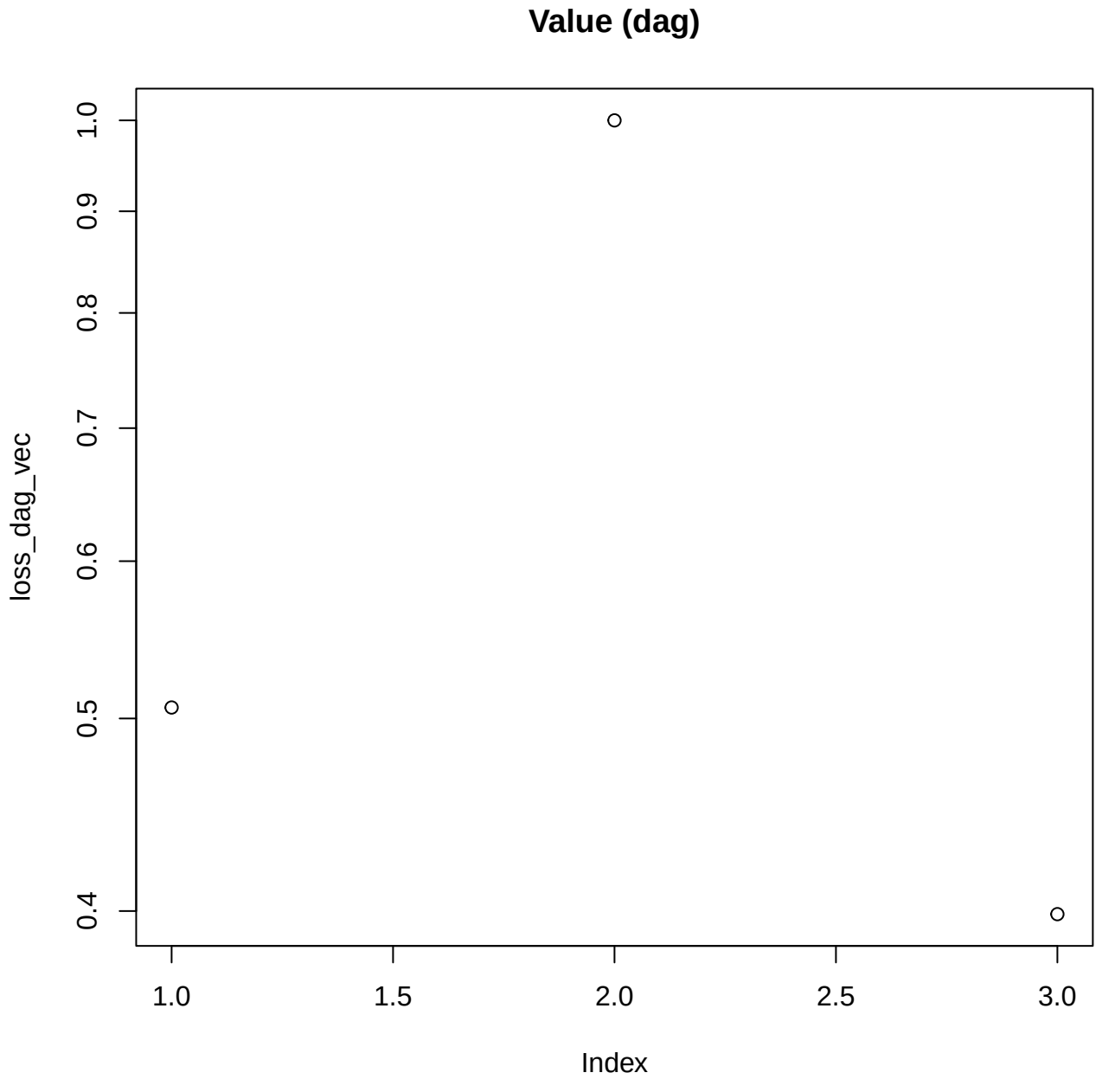
2.5
3.0
Index


Value (dag)
loss_dag_vec
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0.9
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0.7
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0.5
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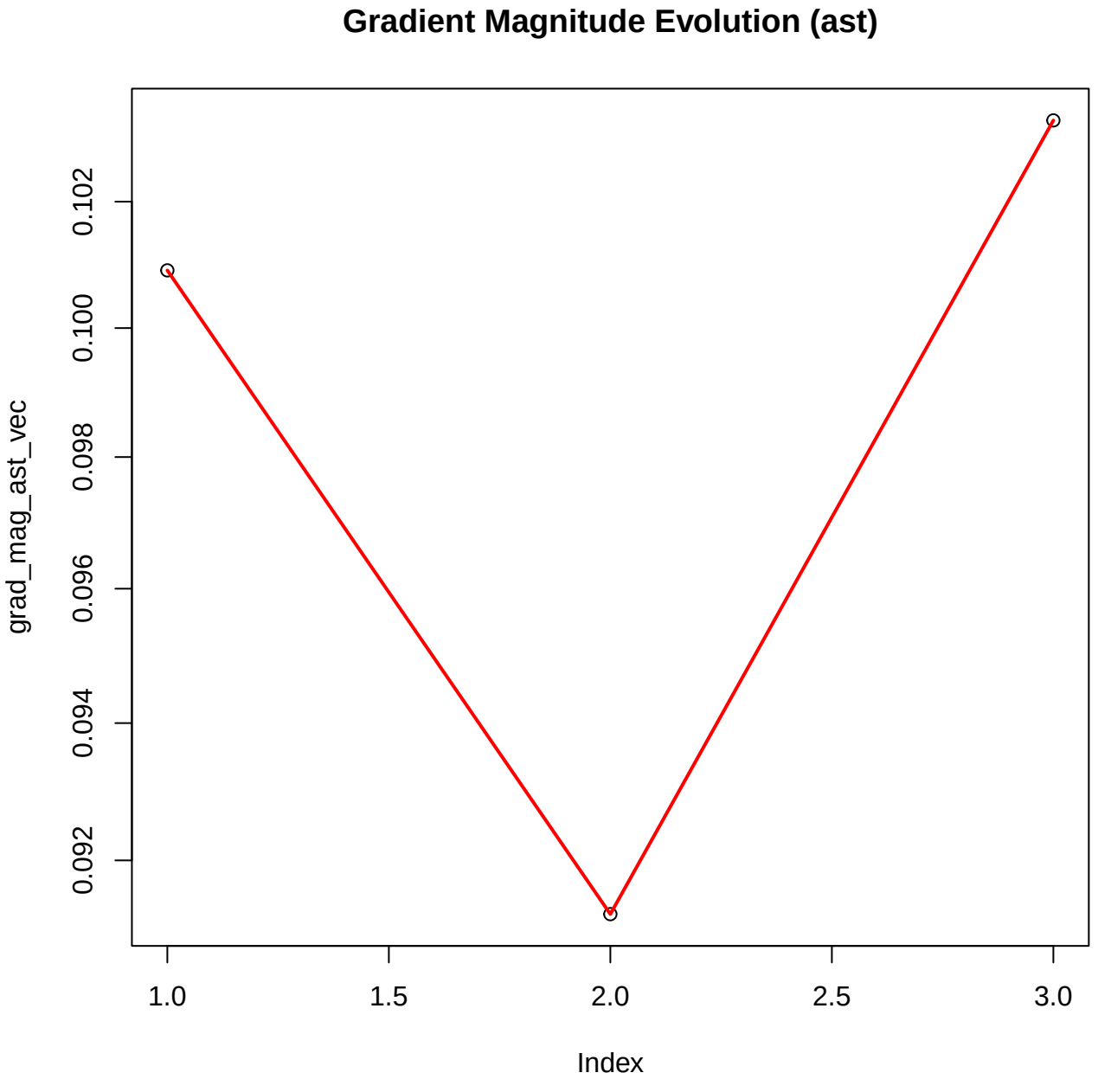
Gradient Magnitude Evolution (ast)
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0.096
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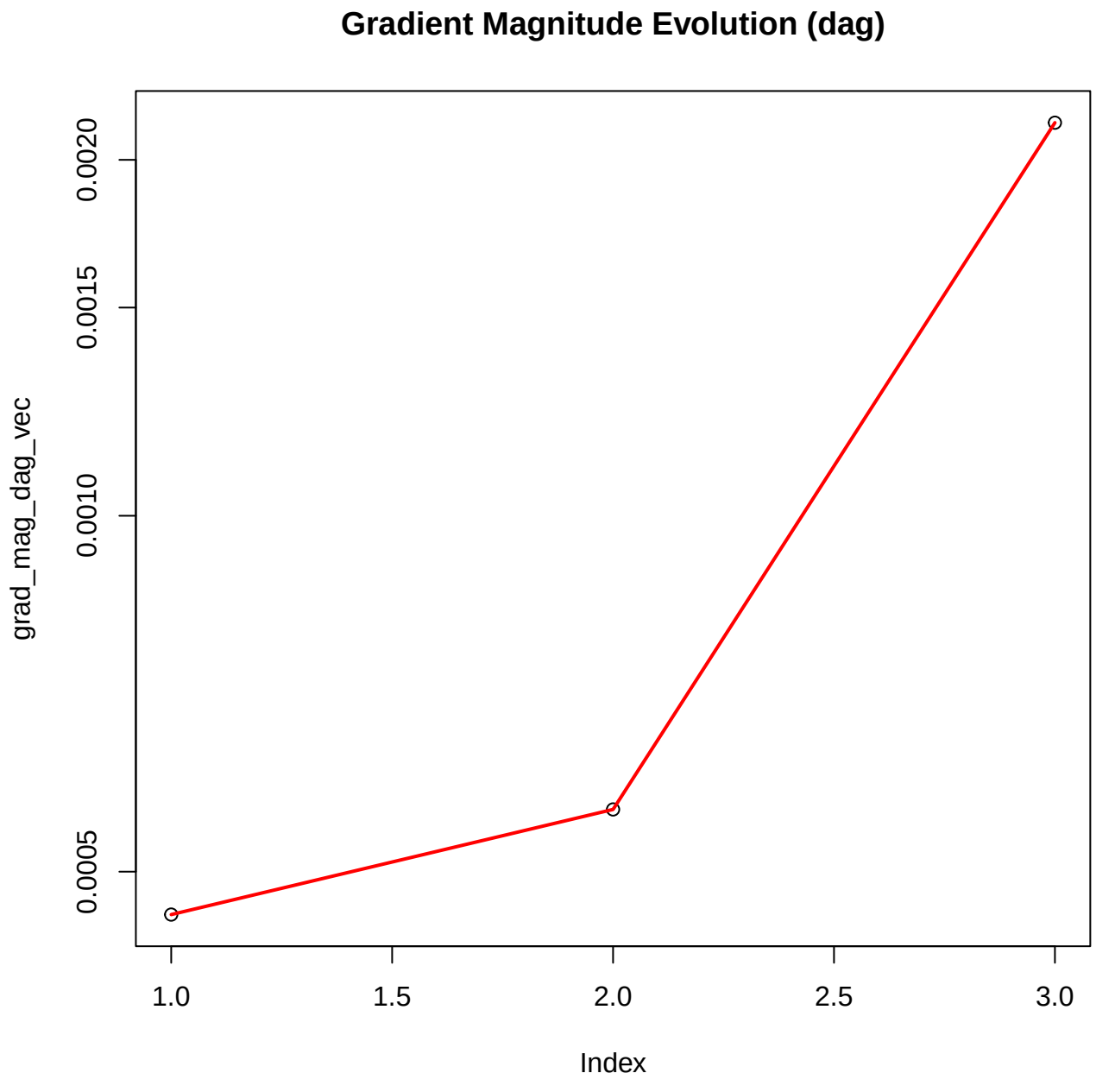
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.0015
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3.0
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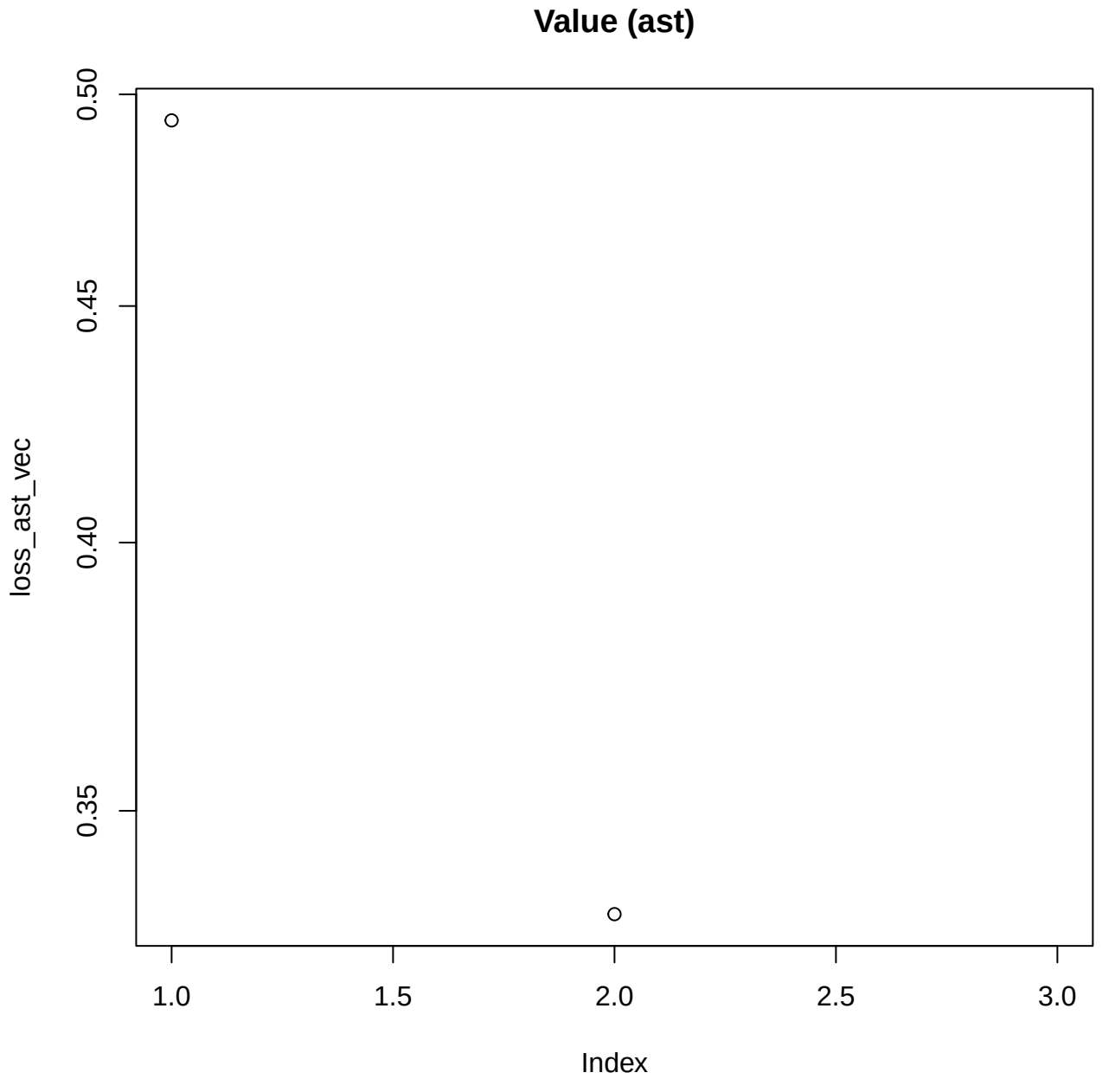
Value (ast)
loss_ast_vec
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0.40
0.35
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1.5
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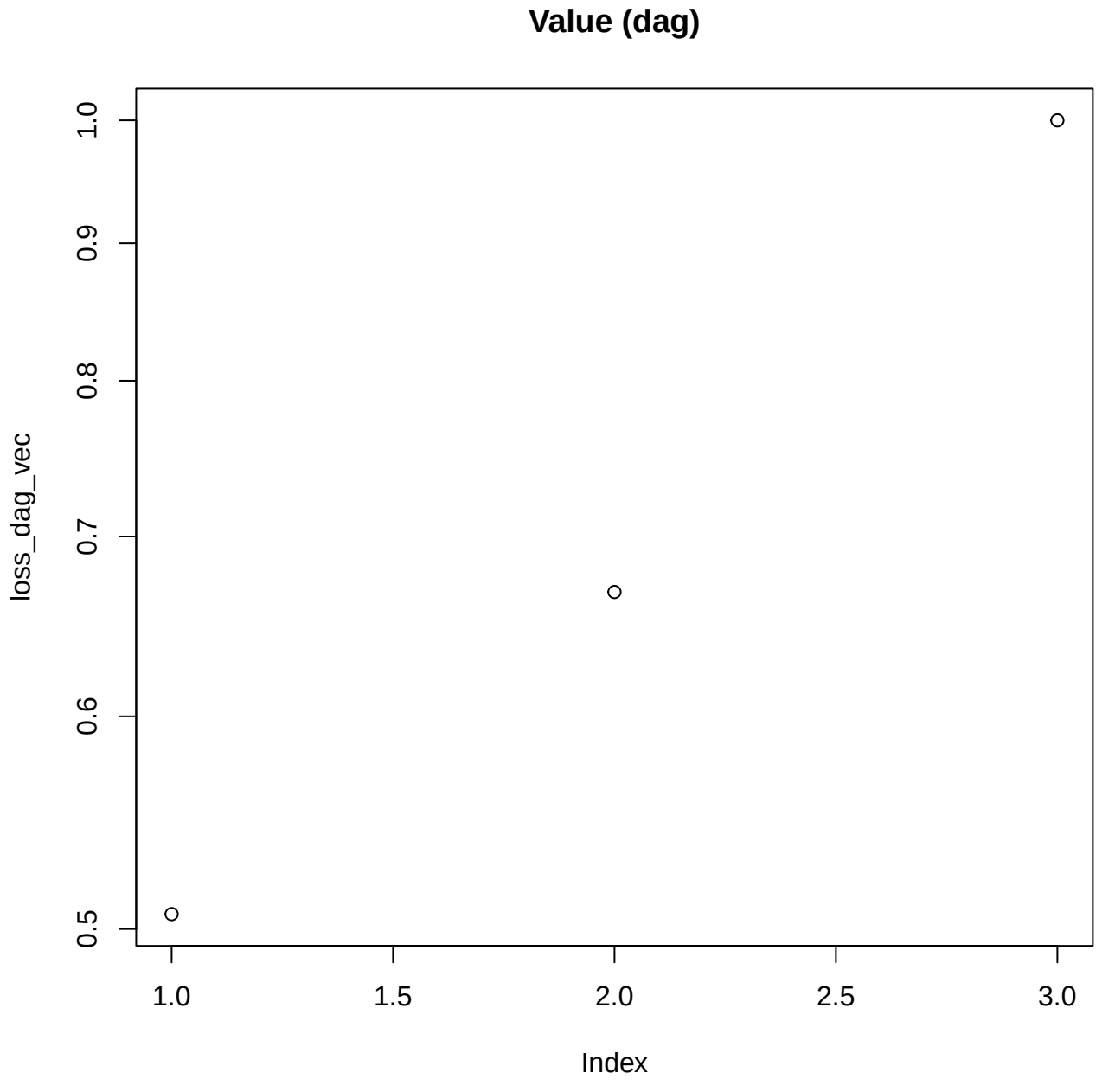
2.5
3.0
Index


Value (dag)
loss_dag_vec
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0.8
0.9
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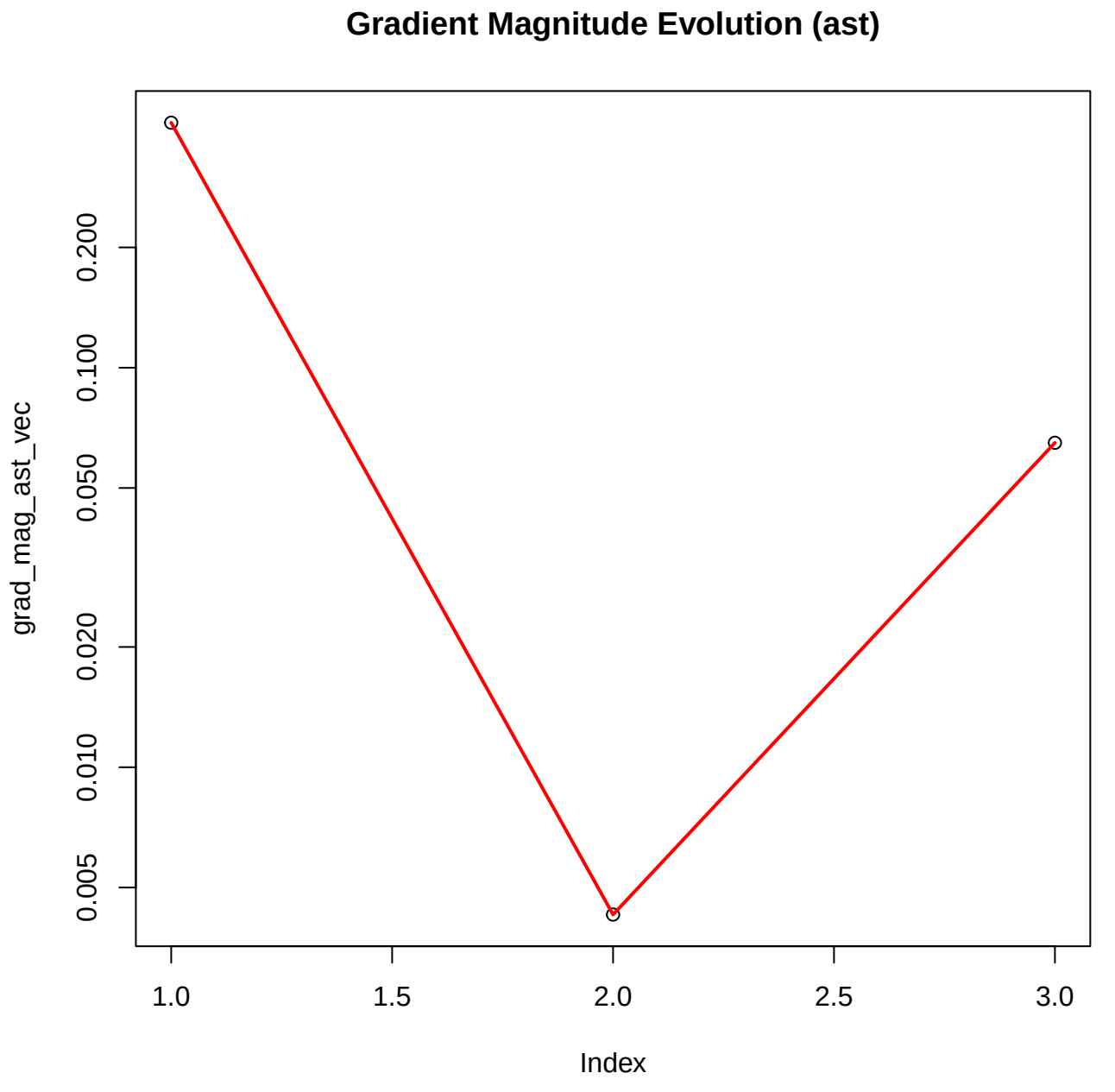
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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3.0
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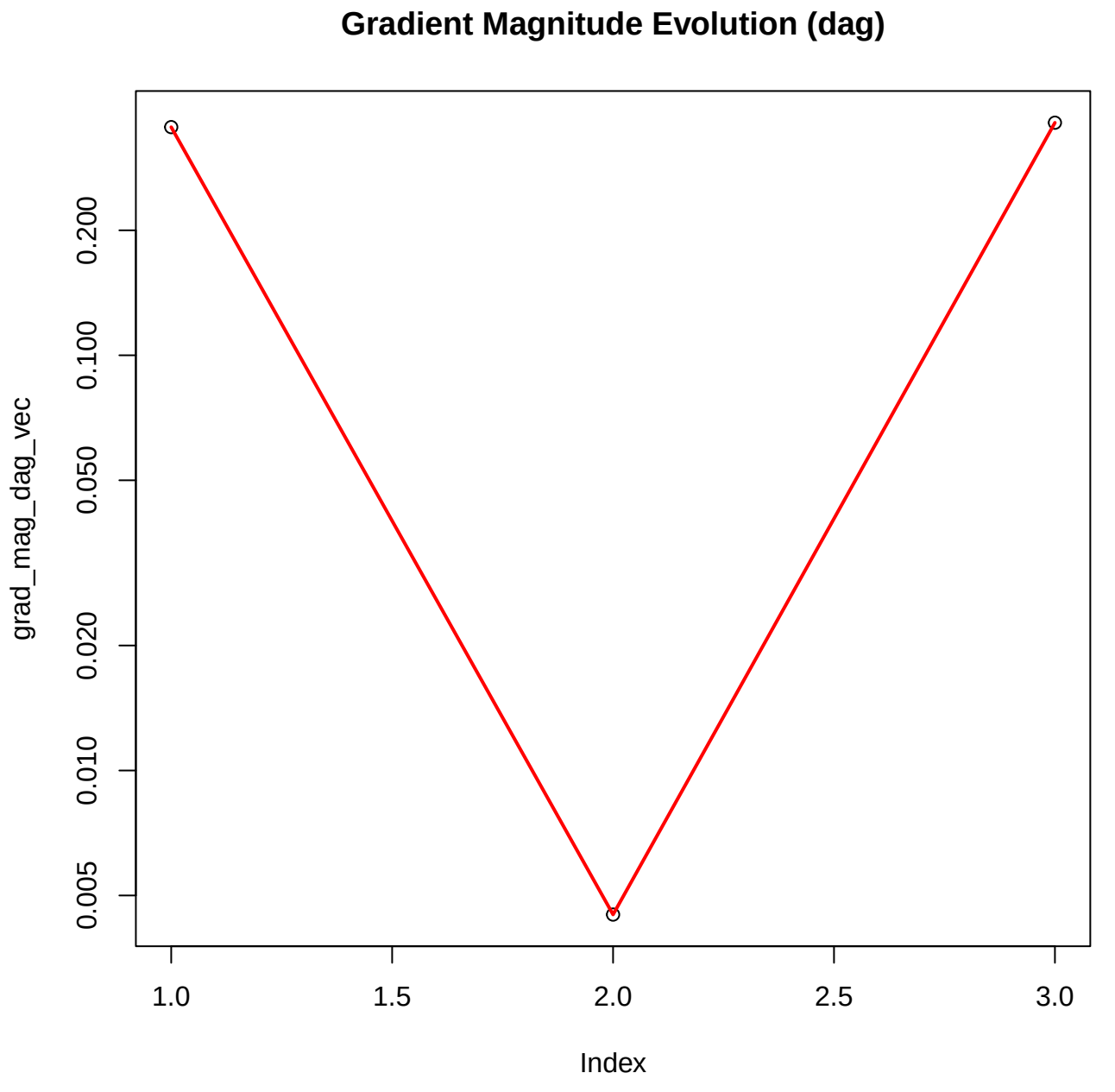
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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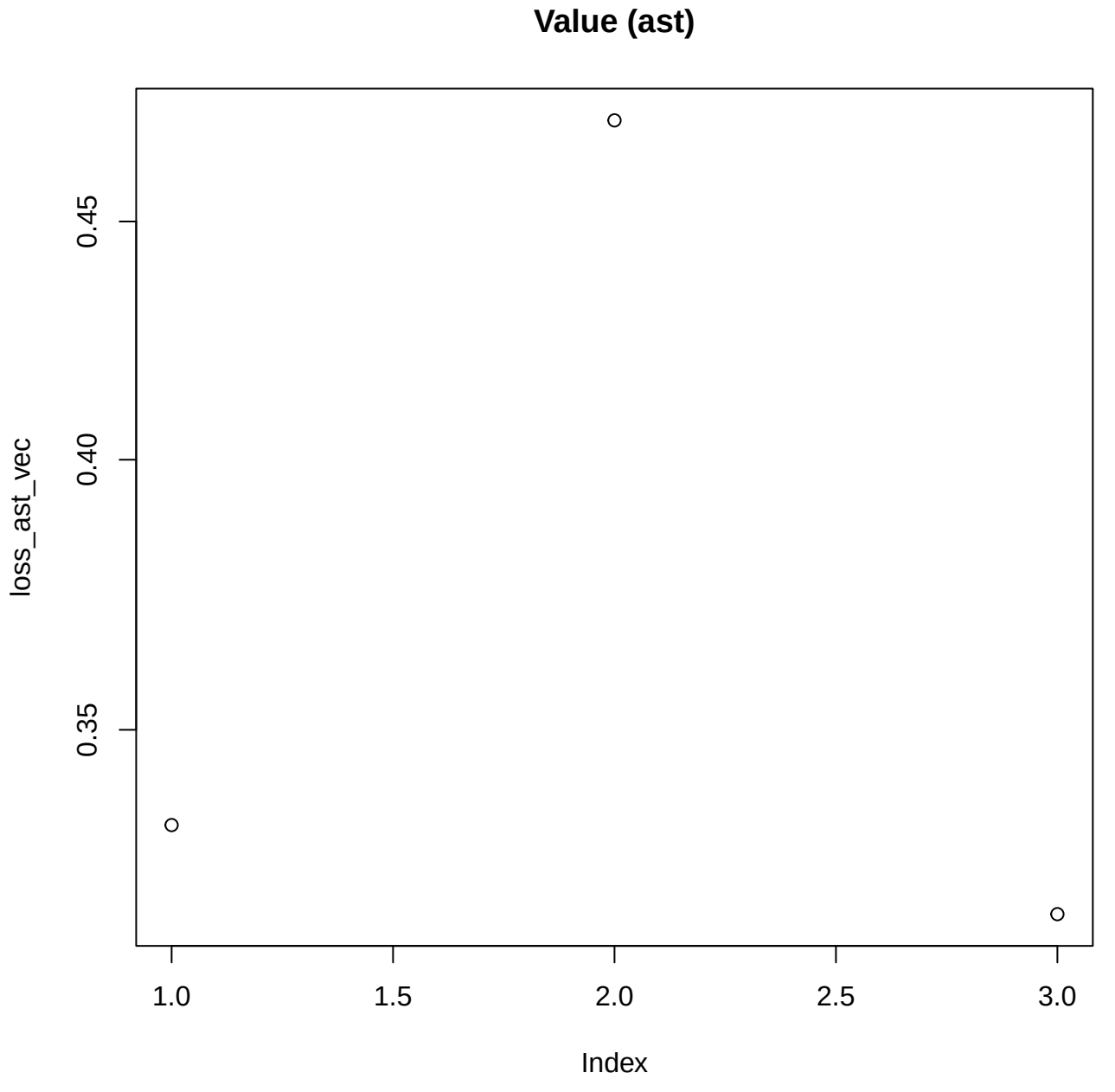
Value (ast)
loss_ast_vec
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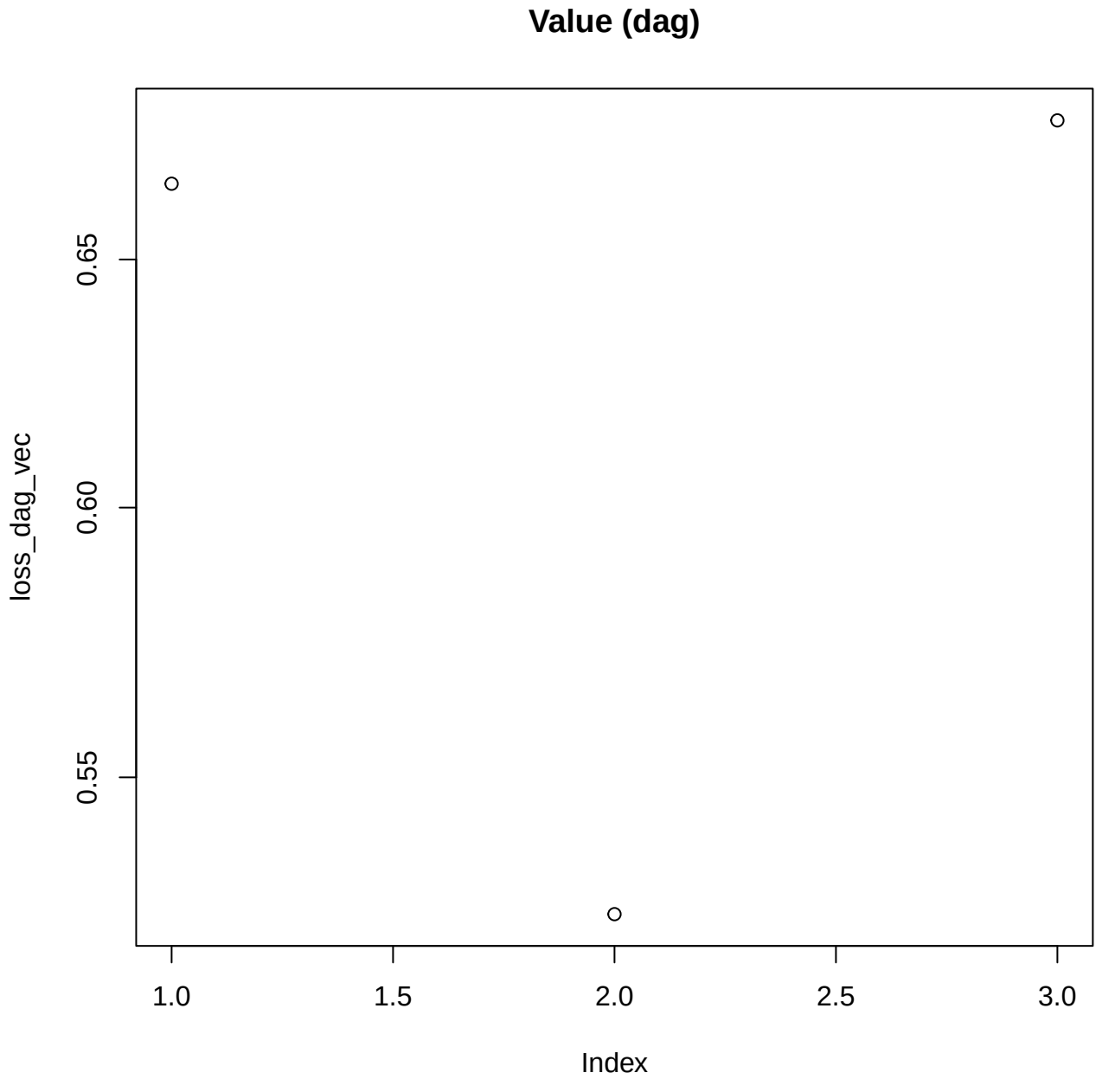
Index


Value (dag)
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2.5
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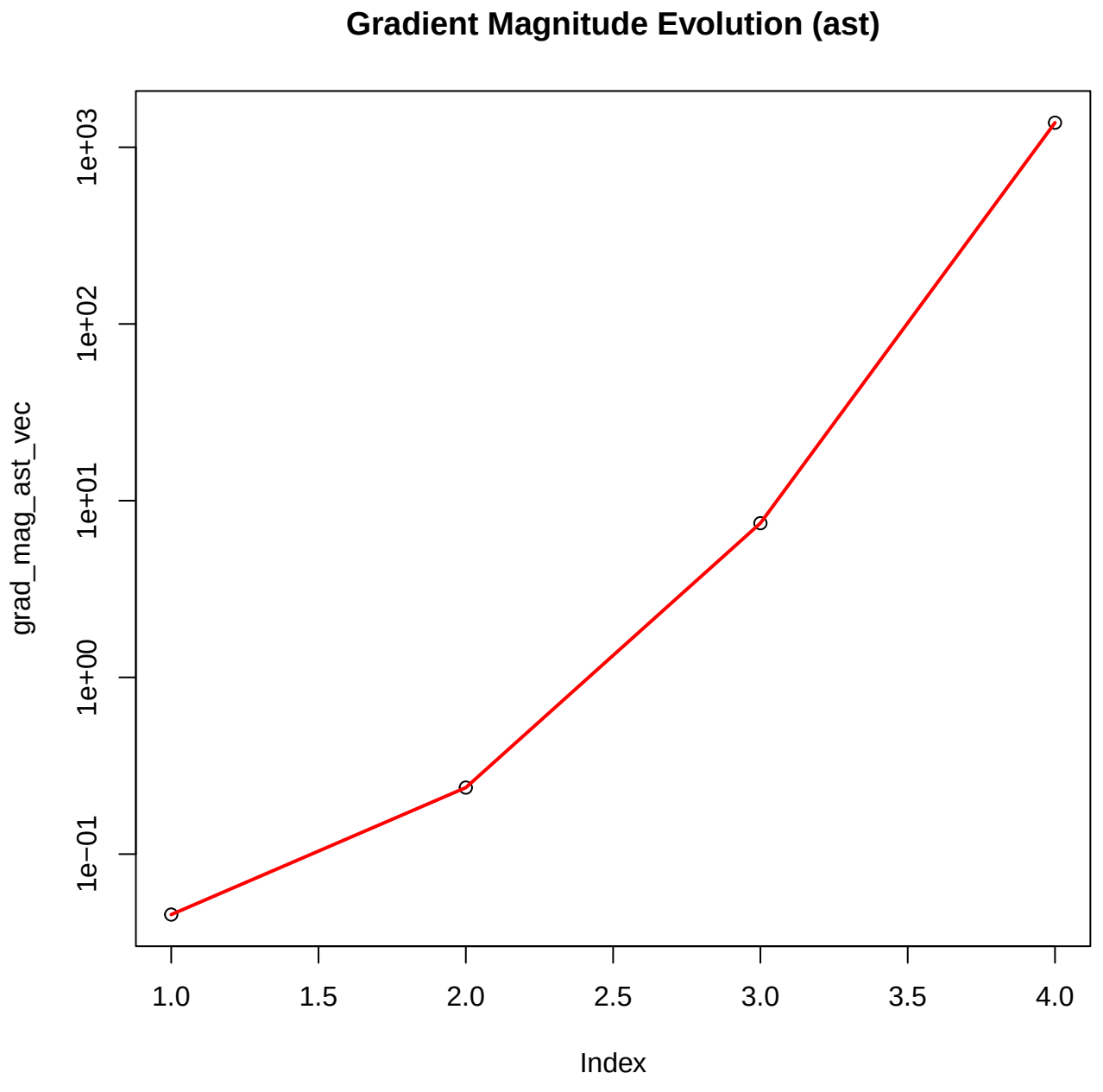
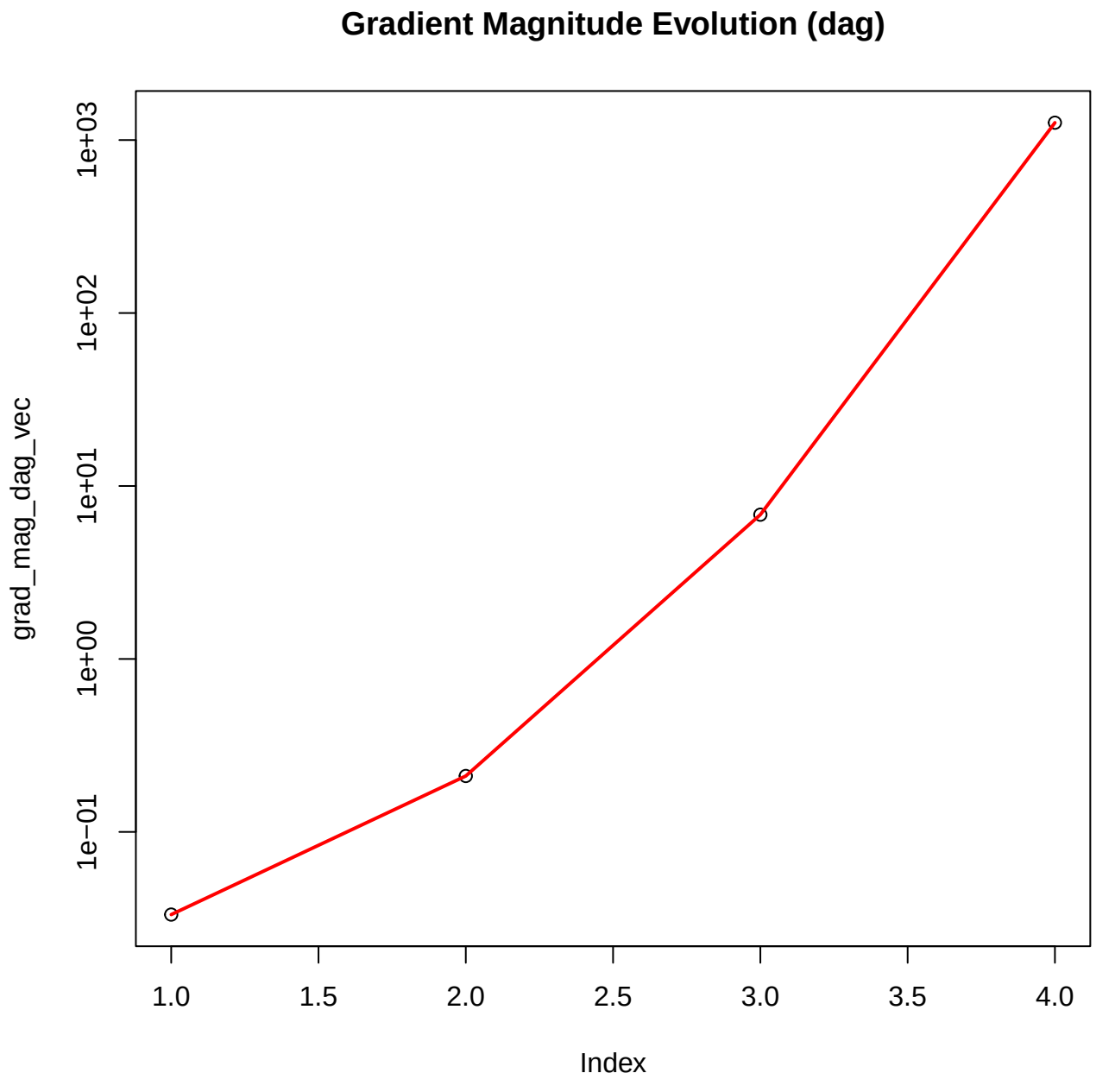
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
1e+00
1e+01
1e+02
1e+03
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1.5
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2.5
3.0
3.5
4.0
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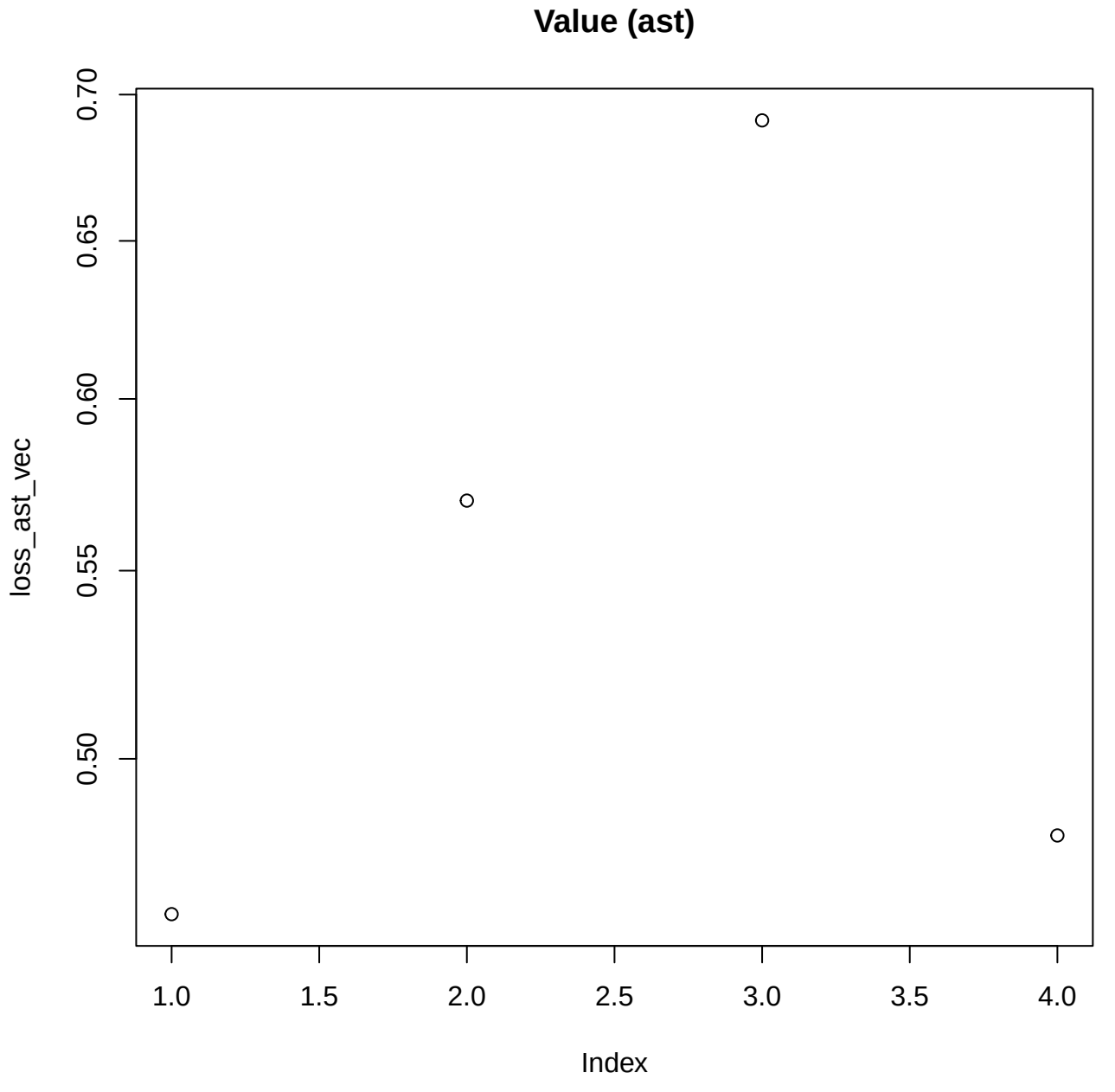
Value (ast)
loss_ast_vec
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2.5
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3.5
4.0
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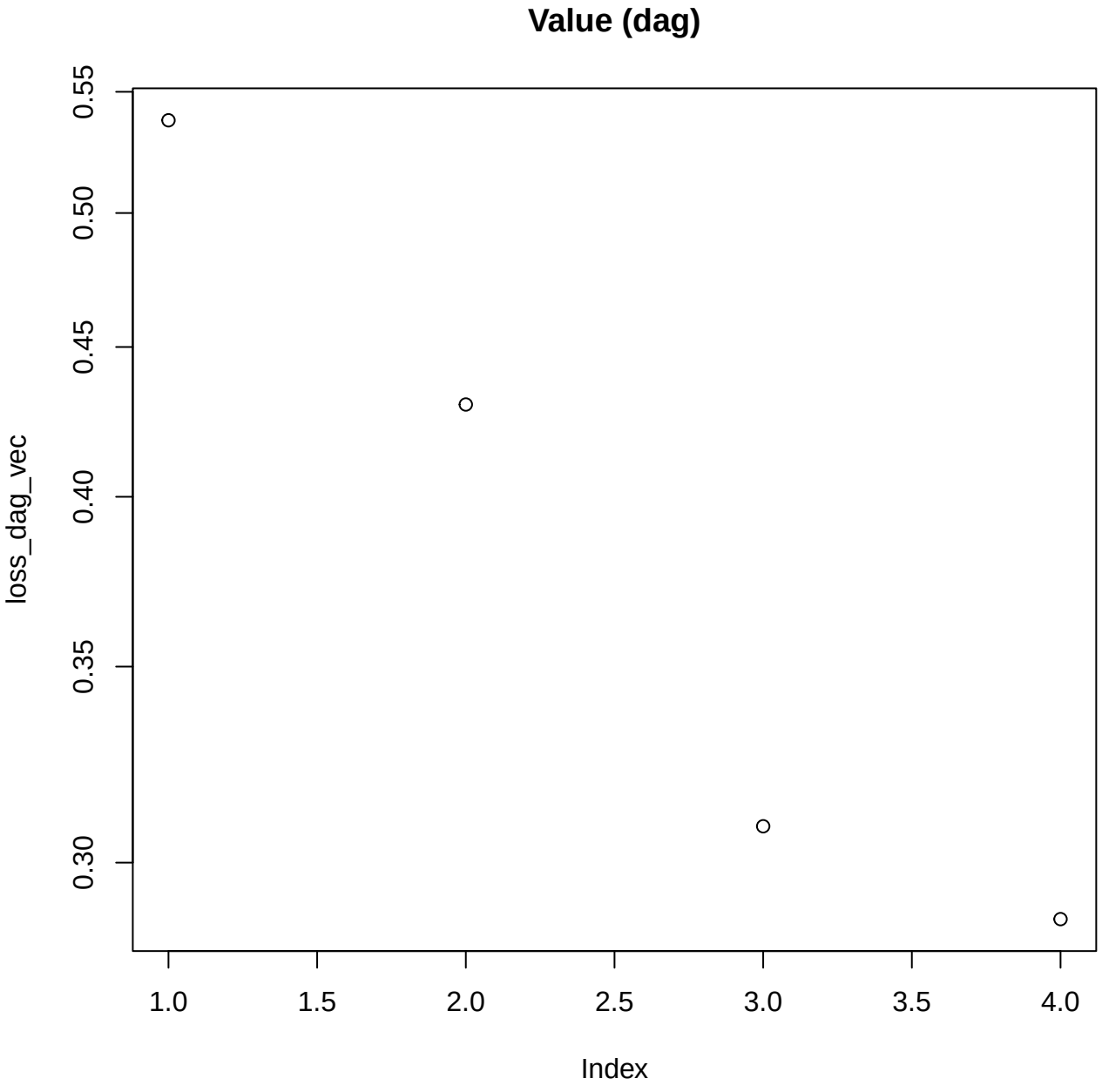
Value (dag)
loss_dag_vec
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0.50
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0.40
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0.30
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1.5
2.0
2.5
3.0
3.5
4.0
Index